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A triangular spectral element method; applications to the incompressible Navier–Stokes equations

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Abstract

Encouraged by the success of spectral elements methods in computational fluid dynamics and p-type finite element methods in structural mechanics we wish to extend these ideas to solving high order polynomial approximations on triangular domains as the next generation of spectral element solvers. We introduce here a complete formulation using a modal basis which has been implemented in a new code *NekTar*. The new basis has the following properties: Jacobi polynomials of mixed weights; semi-orthogonality; hierarchical structure; generalized tensor (*warped*) product; variable order; and a new apex co-ordinate system allowing automated integration with Gaussian quadrature. We have discussed the formulation using a matrix notation which allows for an easy interpretation of the forward and backward transformations. We use this notation to formulate the linear advection and Helmholtz equations in an efficient manner and show that we recover the following properties: well conditioned matrices; asymptotic operation count of $O(N^3)$; scaling $O(N^2)$ of the spectral radius of the weak convective operator; and exponential convergence using polynomials up to $\simeq 40$. Having constructed and numerically analysed these equations we are then able to solve the incompressible Navier–Stokes equations using a high-order splitting scheme. We demonstrate a variety of results showing exponential convergence using both deformed and straight triangular subdomains for both the Stokes and Navier–Stokes problems.

1. Introduction

Over the last two decades there have been two distinct trends in computational fluid dynamics, the first emphasizing the importance of flow simulations in configurations of arbitrary geometric complexity, the second concentrating on high-order accurate algorithms in simpler domains. Developments in both of these areas have been driven by different applications and were dictated by the associated governing equations. Roughly speaking, the first category contains numerical methods appropriate for the solution of the compressible Euler equations describing inviscid, high speed flows, whilst the second category contains high-resolution methods appropriate for the solution of unsteady Navier–Stokes equations to simulate turbulence. The algorithms developed in aerodynamic applications allow for very efficient discretization techniques and unstructured mesh generation strategies based on automatic triangulization/tetrahedrization algorithms [1,2]. Those developed for turbulence simulations have matured greatly and have allowed realistic simulations at Reynolds numbers comparable to wind tunnel experiments [3].

For aerodynamic simulations, the discretizations are typically based on low-order finite elements and finite volume concepts; spectral methods have been used almost exclusively in direct simulations of turbulence [4,5].

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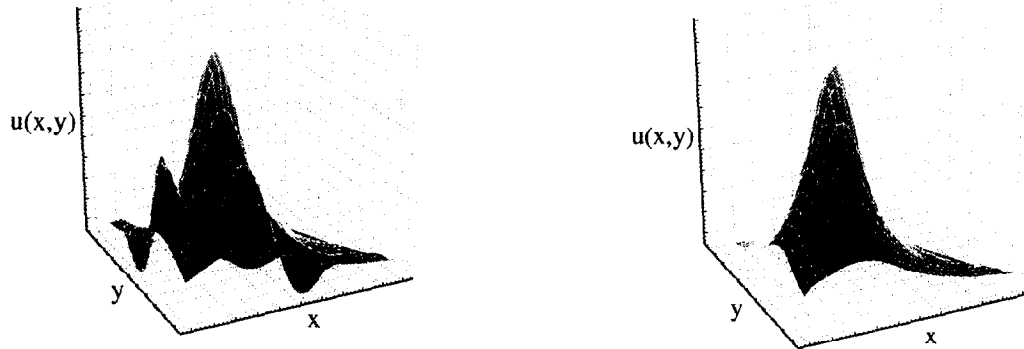


Fig. 1. A Runge type function was represented using Lagrange interpolation (13th order polynomial) on an equispaced mesh in triangular co-ordinated (left) and the triangular basis proposed by Dubiner (right). The Lagrange representation develops large oscillations away from the peak of the function.

More recently, with the interest shifted towards solutions of the Navier–Stokes equations around aerodynamic configurations it has been recognized that higher-order accurate approximations significantly enhance the quality of results; this has motivated several studies into developing high-order reconstruction procedures on unstructured meshes [6–8]. However, the extension of a low-order finite element basis $x^m y^n \{ (m, n) | 0 \leq m, n; m + n \leq 3 \}$ to a higher-order is not a trivial one; for large $(m, n) > 7$ the basis is nearly dependent leading to severely ill-conditioned approximations.

To illustrate how the choice of expansion bases can effect convergence we consider the Runge type function $u(x, y) = 1/(50((x - 0.25)^2 + (y - 0.25)^2) + 1)$ which has a maximum value of 1 at $x = 0.25, y = 0.25$ and decays away from this point like r^2 . The function is represented with similar order expansions using a Lagrange interpolation in triangular co-ordinates as well as the polynomial expansion developed by Dubiner [9]. The Lagrange representation develops oscillations away from the peak instead of decaying monotonically as the Dubiner expansion does as shown in Fig. 1.

Conversely, global spectral methods have been extended to multi-domains (spectral elements) to provide better geometric flexibility [10,11]. Over the past decade spectral element methods have proven to be an efficient and accurate way to solve partial differential equations and in particular the incompressible Navier–Stokes equations [12]. Spectral element methods utilize the fast convergence and good phase properties inherent in singular Sturm–Liouville approximations while allowing the solution domain to include complex geometries. The expense for the improvement is a higher operation count compared with more traditional finite element methods although this may be balanced by the faster (exponential) convergence rate. Therefore, if high accuracy is required these methods prove to be more efficient. However, existing spectral element methods use quadrilateral subdomains, in two dimensions, or hexahedral subdomains, in three dimensions, to discretize the computational domain. This imposes the undesirable requirement of structured grids although flexibility can be somewhat enhanced with non-conforming spectral elements [13,14]. On the other hand, this restricts automated mesh generation and the flexibility in discretization provided by triangular subdomains. Noting the success of triangular finite element and finite volume methods as well as the recent recognition for higher-order triangular elements the obvious extension for spectral methods is to use triangular or tetrahedral subdomains in two or three dimensions, respectively.

Here we present algorithms which allow efficient solution of partial differential equations using triangular elements. This construction can be compared to introducing higher-order bases into finite elements. The degeneration of quadrilateral spectral elements onto triangular domains by collapsing two corners is undesirable since this leads to unacceptable time step restrictions in hyperbolic equation [9]. In recent theoretical work Dubiner [9] has proposed a well conditioned basis for use in triangular domains. The direct (tensor) product of spectral methods on quadrilaterals, responsible for the sum factorization transform, is replaced by a *warped product* which also results in sum factorization and thus the cost of evaluating derivatives or squares of a function is maintained at operation count $O(N^{(d+1)})$ in R^d space as in quadrilateral or hexahedral elements. Dubiner has proposed two bases: The first is completely orthogonal but it is not easy to extend it to form a C^0 continuous basis from the union of triangular subdomains and thus it is inappropriate for spectral element discretizations.

The second *modified* basis is designed to overcome this problem although in doing so the orthogonality between different bases is reduced.

The new algorithm we have developed uses Dubiner's modified basis. Essentially, this basis is made up of *interior* modes, which are zero on the edges of a triangle, and *boundary* modes. The boundary modes can be split into *vertex* and *edge* modes. The vertex modes are simply the linear finite element basis having unit value at one vertex and being zero at the others. The edge modes have non-zero magnitude along one edge only. A three-dimensional basis may be constructed in a similar manner. We have numerically analysed the conditioning of the *mass* and *stiffness* matrices, using this basis, as well as the growth rate of the first- and second-derivative operators. It will be shown that although the conditioning of mass matrix grows relatively quickly it is easily controlled. We are also able to show that the spectrum of the advection operator is bounded by $O(N^2)$ and that the spectrum of the diffusion operator has a growth bounded by $O(N^4)$. These scalings are similar to the standard quadrilateral or hexahedral elements. Examples of these operators are shown in solutions to the linear advection and Helmholtz equations and appropriate fast solvers are introduced. Using a similar construction to that used for the advection and Helmholtz equations we have developed an incompressible Navier–Stokes solver based on a high-order time splitting scheme [15].

This new triangular spectral element method will benefit from the existing algorithms developed for automated triangulization of arbitrary domains. In a sense, it is more general than the fixed-order finite element triangulizations, since both small and large size triangular spectral elements can be employed in the discretization, each consisting of hundreds of modes. Since the algorithm treats modes directly as unknowns, convergence and convergence rates are readily verified by examining the energy of the higher part of the spectrum of computed modes. Similarly, adaptive discretization strategies can be devised based on the same concept of modal contributions and thus efficient *h-p* refinement techniques similar to the ones developed in [16–19] can be incorporated into a triangular spectral element Navier–Stokes solver.

The paper is organized as follows: In Section 2, we present the new triangular, high-order basis and define the warped product, inner products, transforms, and differentiation. In Section 3, we develop discretizations for the advection and Helmholtz equations and examine the eigenspectra of the first- and second-order derivative operators. We also demonstrate through various examples the exponential convergence rate that is obtained using triangular spectral element discretizations. In Section 4, we present the Navier–Stokes algorithm and several benchmark problems and conclude in Section 5.

2. Formulation

2.1. Triangular basis

We wish to define a polynomial basis, denoted by $g_{mn}^{1-2k}(r, s)$, so that we can approximate the function $f(x, y)$ by a C^0 continuous expansion over K triangular subdomains of the form:

$$f(x, y) = \sum_k \sum_m \sum_n \tilde{f}_{mn}^k g_{mn}^{1-2k}(r(x, y), s(x, y)).$$

Here $\tilde{f}_{mn}^k$ is the expansion coefficient for the expansion polynomial, g_{mn}^{1-2k} , in the k^{th} subdomain, (x, y) are the spatial co-ordinates of the function and (r, s) are the local co-ordinates within any given triangle.

The local co-ordinate system (r, s) spans a local space T^2 as shown in Fig. 2 and is mathematically expressed as:

$$T^2 = \{(r, s) | -1 \leq r, s; r + s \leq 0\}.$$

We introduce the local co-ordinate ζ given by

$$\zeta = 2 \frac{(1+r)}{(1-s)} - 1. \quad (1)$$

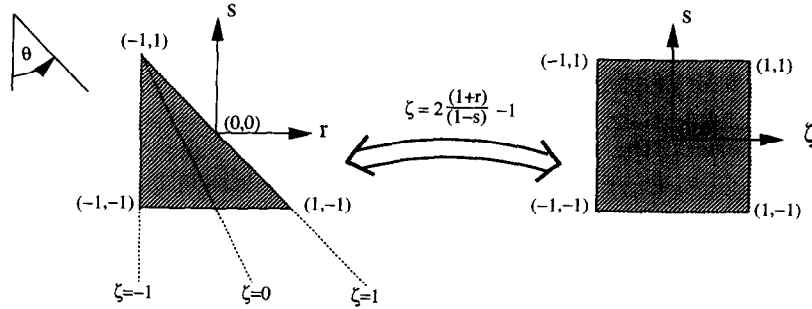


Fig. 2. Definition of the standard triangle.

Within T^2 the co-ordinate ζ has a value of -1 along the line $r = -1$, and a value of 1 , along the line $r + s = 0$, except at the point $r = -1, s = 1$ where ζ is multivalued. To show that ζ is bounded at this point we can consider the substitution $r = -1 + \epsilon \sin \theta, s = 1 - \epsilon \cos \theta$. Here θ is defined in a counter-clockwise manner from the vertical, as indicated in Fig. 2, and ϵ is a small perturbation such that when $\epsilon = 0$ we have $r = -1, s = 1$. Substituting these values into the definition of ζ given by Eq. (1) we can find the limiting behavior of the singularity:

$$\zeta_{-1,1} = 2 \frac{(1 - 1 + \epsilon \sin \theta)}{(1 - 1 + \epsilon \cos \theta)} - 1 = 2 \tan \theta - 1.$$

Since $0 \leq \theta \leq \pi/4$ we know that $0 \leq \tan \theta \leq 1$ and so $-1 \leq \zeta_{-1,1} \leq 1$. Indeed the definition of ζ can be considered as mapping the standard triangle in (r, s) to a bi-unit rectangle in (ζ, s) .

Dubiner [9] proposed two types of polynomial expansion bases for use on triangular domains. The first expansion basis is orthogonal in the Legendre inner product but cannot easily be extended to form a C^0 continuous basis from the union of multiple triangular domains or *elements*. The second expansion basis is only partially orthogonal. However, it can be extended to form a C^0 continuous basis from such a union. Dubiner originally defined these bases in the unit space $T^2 = \{(r, s) | 0 \leq r, s; r + s \leq 1\}$ but here they have been recast into the standard triangle as given above.

2.1.1. Dubiner's orthogonal basis

Let us denote $P_n^{\alpha, \beta}(x)$ as the n^{th} -order Jacobi polynomial in the $[-1, 1]$ interval with the orthogonality relationship,

$$\int_{-1}^1 P_m^{\alpha, \beta}(x) P_n^{\alpha, \beta}(x) (1-x)^\alpha (1+x)^\beta dx = \delta_n^m, \quad (2)$$

where δ_n^m is the Kronecker-delta, then Dubiner's orthogonal expansion basis is given by:

$$g_{mn}^{1-2}(r, s) = P_m^{0,0} \left(2 \frac{(1+r)}{(1-s)} - 1 \right) (1-s)^m P_n^{2m+1,0}(s).$$

We note that this is a polynomial in (r, s) since the $(1-s)^m$ factor acting on the $P_m^{0,0} (2(1+r)/(1-s) - 1)$ Jacobi polynomial produces an m^{th} -order polynomial in r . This $(1-s)^m$ *scaling factor* is very important in scaling the first-order weak convective operator (see Section 3.1.2). The basis can also be expressed as the product of two polynomials in (ζ, s) space i.e.

$$g_{mn}^{1-2}(r, s) = g_m^1 \left(2 \frac{(1+r)}{(1-s)} - 1 \right) \cdot g_{mn}^2(s) = g_m^1(\zeta) \cdot g_{mn}^2(s)$$

where

$$g_m^1(\zeta) = P_m^{0,0}(\zeta), \quad g_{mn}^2(s) = (1-s)^m P_n^{2m+1,0}(s).$$

This property is particularly important when evaluating integrals involving $g_{mn}(r, s)$ over T^2 since it is possible to write the surface integral as the product of two line integrals as explained in Section 2.2.1. Dubiner

refers to this property as a *warped product*. A basis is said to have a warped product relative to inner product $(\cdot, \cdot)$ if and only if:

$$(1, 1)(\overset{1}{f}\overset{2}{f}, \overset{1}{g}\overset{2}{g}) = (\overset{1}{f}, \overset{1}{g})(\overset{2}{f}, \overset{2}{g}).$$

The significance of this property is that the inner product between two polynomial bases $\overset{1}{f}\overset{2}{f}$ and $\overset{1}{g}\overset{2}{g}$ which both span a two-dimensional space can be expressed as the product of two one-dimensional inner products, $(\overset{1}{f}, \overset{1}{g})$ and $(\overset{2}{f}, \overset{2}{g})$ multiplied by a constant. This property is explained further in Section 2.2.1. Finally, we note that the basis is complete in a polynomial space G_M where G_M is defined by:

$$G_M = \text{Span}\{r^m, s^n\}_{(mn) \in M}^{1-2}$$

where

$$M^{1-2} = \{(m, n) | 0 \leq m, n; m < M, m + n < N\}, \quad M \leq N.$$

Also, $\overset{1-2}{g}_{mn}(r, s)$ is orthogonal in the Legendre internal product defined by

$$(\overset{1-2}{g}_{mn}(r, s), \overset{1-2}{g}_{pq}(r, s)) = \iint_{T^2} \overset{1-2}{g}_{mn}(r, s) \overset{1-2}{g}_{pq}(r, s) dr ds = \frac{1}{2} \delta_p^m \delta_q^n.$$

To demonstrate this result we can perform a co-ordinate transformation from (r, s) to (ζ, s) . Since

$$\zeta = 2 \frac{(1+r)}{(1-s)} - 1 \Rightarrow J \equiv \frac{\partial(r, s)}{\partial(\zeta, s)} = \begin{vmatrix} \frac{(1-s)}{2} & \frac{-(1+\zeta)}{2} \\ 0 & 1 \end{vmatrix} = \frac{(1-s)}{2}$$

the inner product becomes:

$$(\overset{1}{g}_m \overset{2}{g}_{mn}, \overset{1}{g}_p \overset{2}{g}_{pq}) = \frac{1}{2} \int_{-1}^1 P_m^{0,0} P_p^{0,0} d\zeta \int_{-1}^1 P_n^{2m+1,0} P_q^{2p+1,0} (1-s)^{m+p+1} ds.$$

Now from the definition of the Jacobi polynomial, $P_n^{\alpha,\beta}(x)$, we see from relation (2) that the first integral is zero if $m \neq p$. When $m = p$ the second integral is zero if $n \neq q$. This can be appreciated as the function $(1-s)^{m+p+1}$ is the weight function in relation (2) and so the orthogonality of the base $\overset{1-2}{g}_{mn}(r, s)$ is verified.

This basis would be ideal except that we want to discretize the solution domain into triangular subdomains, or *elements*, and enforce a C^0 continuity between each subdomain. C^0 continuity is not easily enforced but might be achieved by a constraint on all modes. A constraint of this sort would introduce a non-diagonal contribution to the matrices thereby destroying the most desirable property of the expansion. To overcome this problem, Dubiner suggested the following modified triangular basis.

2.1.2. Dubiner's modified triangular basis

The basis is now split into *interior* and *boundary* modes where all interior modes are zero on the triangle boundary similar to the *bubble* modes used in p-type finite elements [18,20]. The boundary modes can further be described in terms of *vertex* and *edge* modes. The vertex modes vary linearly from a unit value at one vertex to zero at the others. The edge modes only have magnitude along one edge and are zero at all other vertices and edges. Using the notation given in Fig. 3 and recalling that $P_n^{\alpha,\beta}(x)$ refers to the Jacobi polynomial we can write the modified basis as follows:

- Interior modes ($2 \leq m; 1 \leq n, m < M; m + n < N$)

$$\overset{1-2}{g}_{mn}^{\text{interior}} = \left(\frac{1+\zeta}{2}\right) \left(\frac{1-\zeta}{2}\right) P_{m-2}^{1,1}(\zeta) \cdot \left(\frac{1-s}{2}\right)^m \left(\frac{1+s}{2}\right) P_{n-1}^{2m-1,1}(s);$$

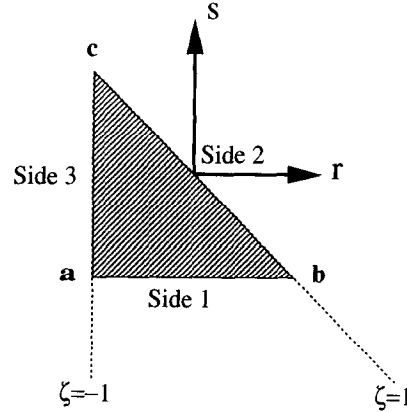


Fig. 3. Definition of standard triangle for modified basis.

- Edge modes ($2 \leq m$; $1 \leq n$, $m < M$; $m + n < N$)

$$g_{m0}^{1-2\text{side}-1} = \left(\frac{1+\zeta}{2}\right) \left(\frac{1-\zeta}{2}\right) P_{m-2}^{1,1}(\zeta) \cdot \left(\frac{1-s}{2}\right)^m$$

$$g_{1n}^{1-2\text{side}-2} = \left(\frac{1+\zeta}{2}\right) \cdot \left(\frac{1-s}{2}\right) \left(\frac{1+s}{2}\right) P_{n-1}^{1,1}(s)$$

$$g_{1n}^{1-2\text{side}-3} = \left(\frac{1-\zeta}{2}\right) \cdot \left(\frac{1-s}{2}\right) \left(\frac{1+s}{2}\right) P_{n-1}^{1,1}(s);$$

- Vertex modes

$$g_{1-2\text{vert}-a} = \left(\frac{1-\zeta}{2}\right) \cdot \left(\frac{1-s}{2}\right)$$

$$g_{1-2\text{vert}-b} = \left(\frac{1+\zeta}{2}\right) \cdot \left(\frac{1-s}{2}\right)$$

$$g_{1-2\text{vert}-c} = 1 \cdot \left(\frac{1+s}{2}\right).$$

In order to enforce C^0 continuity between adjacent triangular elements we require that $N = M$. The number of modes to be used in a basis is therefore dictated by the value of M . An interpretation of M is that the maximum order of the polynomial along any edge is $(M - 1)$. For example when $M = 2$ we only have 3 vertex modes which gives us a linear finite elements basis that has a maximum polynomial order of $M - 1 = 1$ along any edge. This basis is once again complete in the polynomial space G_M as defined in the previous section. The interior modes are similar to the orthogonal basis except that there is now a $\left(\frac{1+\zeta}{2}\right)\left(\frac{1-\zeta}{2}\right)$ factor in both the r and s dependent polynomials. It is this factor which ensures that the interior modes are zero on the boundaries. To help maintain orthogonality between the modes the Jacobi polynomials in r now have $\alpha = 1, \beta = 1$ which is beneficial as this makes $(1+r)(1-r)$ the weight function in the inner product. The polynomial $(1-z)(1+z)P_n^{1,1}(z)$ is related to the integral of the Legendre polynomial $P_n^{0,0}(z)$ by a constant factor [21] i.e. $2n \int_{-1}^z P_n^{0,0}(s) ds = -(1-z)(1+z)P_{n-1}^{1,1}(z)$. This is interesting to note since the integral of the Legendre polynomial is used in the p-type finite element community [18,20]. The edge modes now have exactly the same shape for $n = m - 1$. This is easily seen once it is appreciated that along side 1, $s = -1$; along side 2, $\zeta = 1$ and along side 3, $\zeta = -1$. Since the vertex modes are the same as the linear finite element basis it can be appreciated that by matching the vertex and edge modes multiple triangular domains can be joined together in a contiguous fashion. The only complication is that some edge modes may be the negative of the adjacent matching mode. This point is illustrated in Fig. 4 which shows all the modes for $M = 5$.

At this order there are three interior modes and three edge modes on each side. Along the edges one a quadratic form, one a cubic form and last one has a quartic form. It can be appreciated that the cubic form mode on side 1 is the negative of the cubic form mode on side 2 whilst all of the quadratic and quartic form

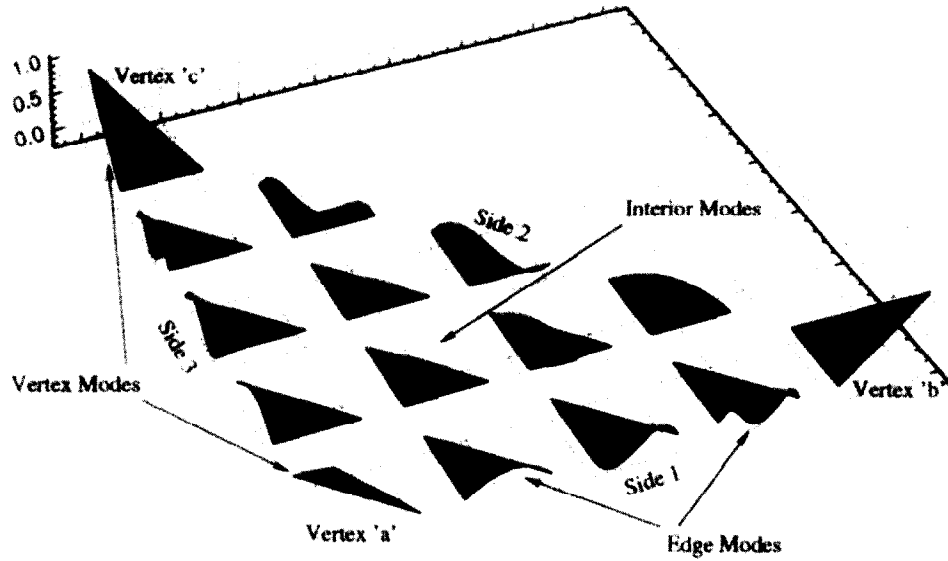


Fig. 4. Shape of all modes for the case $M = 5$. All interior modes are zero on the boundaries whilst the edge modes are zero on adjacent edges and the vertex modes are zero on adjacent vertices.

modes are of the same sign. So the connectivity condition is enforced by initially stating that all triangular elements must be defined in an counter clockwise sense; then if a side 1 meets a side 2 or like sides meet (e.g. side 3 with side 3) the odd modes of one of the triangular elements will be negated. This condition is not as difficult as it may seem to implement since only the information that a mode must be negated need be stored, all local operations can be performed without knowledge of how the modes are connected.

As mentioned earlier not all the modes are orthogonal in this basis. To see what orthogonality does exist we can consider the inner product of two sets of the interior modes over the T^2 space.

$$\left(g_{mn}^{1-2^{interior}}, g_{pq}^{1-2^{interior}} \right)_{T^2} = \int_{-1}^1 \left(\frac{1+\zeta}{2} \right)^2 \left(\frac{1-\zeta}{2} \right)^2 P_{m-2}^{1,1} P_{p-2}^{1,1} d\zeta \int_{-1}^1 \left(\frac{1+s}{2} \right)^2 \left(\frac{1-s}{2} \right)^{m+p} P_{n-1}^{2m-1,1} P_{q-1}^{2p-1,1} \frac{(1-s)}{2} ds \quad (3)$$

where $P_m^{1,1}(z)$ is orthogonal in the inner product:

$$(P_m^{1,1}(z), v(z))_{1,1} = \int_{-1}^1 (1+z)(1-z) P_m^{1,1}(z) v(z) dz.$$

The value of the integral will be zero if $v(z)$ is a polynomial of order $(m-1)$ or less, since in this case we can recast $v(z)$ into $v(z) = \sum_{i=0}^{m-1} \hat{v}_i P_i^{1,1}(z)$. We can write the first integral of (3) as:

$$\int_{-1}^1 (1+\zeta)(1-\zeta) P_{m-2}^{1,1}(\zeta) f_p(\zeta) d\zeta, \quad \text{where } f_p(\zeta) = \frac{1}{16} (1+\zeta)(1-\zeta) P_{p-2}^{1,1}(\zeta).$$

So this integral will be zero if $(m-2) > p$. Similarly it will also be zero if $(p-2) > m$ and both expressions can be summarized as $|m-p| > 2$. The practical significance of this orthogonality can be appreciated when evaluating

the mass matrix. Considering the interior mode component of the mass matrix, $(g_{mn}^{1-2^{interior}}, g_{pq}^{1-2^{interior}})$, the above orthogonality result says that the matrix will have an upper bandwidth of $(M-3) + (M-4) + (M-5) - 1 \approx 3M$. However, if we now consider the second integral of Eq. (3) for the case when $|m-p| = 2$ it can be shown that the upper bandwidth can be reduced to $(M-3) + (M-4) + 1 \approx 2M$. Considering the case where $(m-p) = 2$, so $p = m-2$, the second integral of (3) can be written as:

$$\int_{-1}^1 (1+s)(1-s)^{2m-1} P_{n-1}^{2m-1,1} f_q(s) ds, \quad \text{where } f_q(s) = \frac{1}{2^{2m+1}} P_{q-1}^{2m-5,1} ds.$$

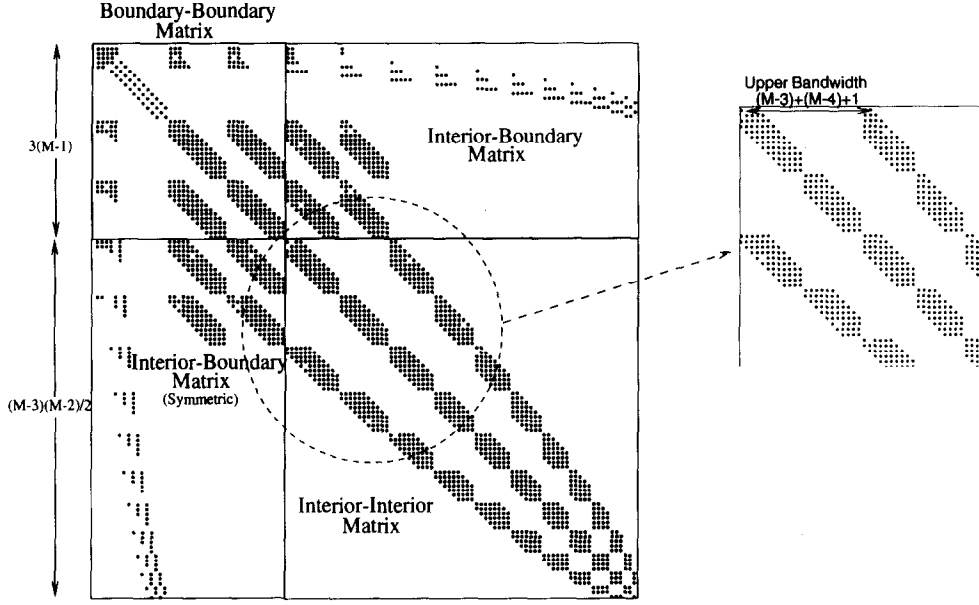


Fig. 5. The structure of the mass matrix on the standard triangle for $M = 15$. The matrix is symmetric; the boundary modes are listed first followed by the interior modes which have a banded structure. The matrix rank is $M(M + 1)/2$.

This integral is clearly zero if $(n - 1) > q$ and similarly it can be shown that the integral is also zero for the case where $(p - m) = 2$ and $(q - 1) > n$. This leads to the result that the diagonal of the mass matrix will be zero if $|m - p| > 2$. If $|m - p| = 2$ then it is zero if $|n - q| > 1$ which means the mass matrix has an upper bandwidth of $(M - 3) + (M - 4) + 1$ where the $(M - 3) + (M - 4)$ contribution is from the orthogonality between the m, p part of the modes and the extra 1 is from the orthogonality of the n, q part of the modes. Finally, it should also be noted that there is similar orthogonality between the interior and edge modes which introduces further sparsity into the mass matrix structure. This structure is clearly evident in Fig. 5 which shows the form of the mass matrix for $M = 15$.

2.2. Integration, notation, transformations and differentiation

2.2.1. Integration

Since the bases are polynomials in (ζ, s) , integration over T^2 can be evaluated exactly by using Gaussian quadrature. We know that the Jacobian introduces a factor of $(1 - s)/2$ into the integral and this factor can be incorporated into the quadrature weights by using the weights corresponding to the zeros of the $P^{1,0}(z)$ Jacobi polynomial, i.e.

$$\int_{-1}^1 f(z) \frac{(1-z)}{2} dz = \sum_{i=0}^{N-1} f(z_i^{1,0}) \frac{w_i^{1,0}}{2} = \sum_{i=0}^{N-1} f(z_i^{1,0}) \hat{w}_i^{1,0}$$

where $z_i^{1,0}$ and $w_i^{1,0}$ are the zeros and weights related to the $P^{1,0}(z)$ Jacobi polynomial, respectively, and $\hat{w}_i^{1,0} = w_i^{1,0}/2$. This is exact if $f(z)$ is a polynomial of order $(2N - 1)$ and Gauss type quadrature is used, and for polynomials of order $(2N - 3)$ if Gauss-Lobatto type quadrature is used. Therefore, an arbitrary polynomial function $u(r, s)$ can be integrated over the standard triangle in the following discrete manner:

$$\int \int_{T^2} u(r, s) dr ds = \int_{-1}^1 \int_{-1}^1 u(\zeta, s) \frac{(1-s)}{2} d\zeta ds = \sum_{i=0}^{q_\zeta-1} \sum_{j=0}^{q_s-1} u(\zeta_i^{0,0}, s_j^{1,0}) w_i^{0,0} \hat{w}_j^{1,0},$$

where q_ζ, q_s are the number of quadrature points in the ζ and s directions, respectively. Here the $\zeta_i^{0,0}$ and $w_i^{0,0}$ refer to the zeros and weights related to the $P^{0,0}(\zeta)$ Jacobi polynomial which is the Legendre polynomial.

Clearly this is a $q_\zeta \cdot q_s$ operation and is exact for a polynomial of order $(2q_\zeta - 1)$ in ζ and $(2q_s - 1)$ in s if Gauss points are used. However, this $q_\zeta \cdot q_s$ operation count is reduced to a factor of $(q_\zeta + q_s)$ when integrating the basis $^{1-2}g_{mn}(r, s)$ due to the warped product nature of the basis in the (ζ, s) space so:

$$\begin{aligned} \int \int_{T^2} ^{1-2}g_{mn} &= \int_{-1}^1 \int_{-1}^1 g_m^1(\zeta) g_{mn}^2(s) \frac{(1-s)}{2} d\zeta ds \\ &= \int_{-1}^1 g_m^1(\zeta) d\zeta \int_{-1}^1 g_{mn}^2(s) \frac{(1-s)}{2} ds. \end{aligned}$$

The last integral can be evaluated in a discrete fashion as:

$$\int_{-1}^1 g_m^1(\zeta) d\zeta \int_{-1}^1 g_{mn}^2(s) \frac{(1-s)}{2} ds = \sum_{i=0}^{q_\zeta-1} g_m^1(\zeta_i^{0,0}) w_i^{0,0} \sum_{j=0}^{q_s-1} g_{mn}^2(s_j^{1,0}) \hat{w}_j^{1,0},$$

which involves $(q_\zeta + q_s)$ operations. Since the $g_m^1(\zeta)$ polynomial has a maximum order of $(M - 1)$ and for reasons to be shown later we wish to be able to evaluate the inner product of $(^{1-2}g_{mn}, ^{1-2}g_{pq})$ we require that $2q_\zeta - 3 > 2(M - 1)$ when using Gauss–Lobatto integration. Similarly the $g_{mn}^2(s)$ polynomial also has a maximum order of $(M - 1)$ so we choose $q_\zeta = q_s = M + 1$ in order to evaluate all inner products exactly. This can be slightly improved by using the Gauss–Radau point in the s direction with only one quadrature point at $s = -1$. This is also preferable when differentiating since no evaluations are necessary at the singular point (see Section 2.2.3). As a final point we note that integration over an arbitrary triangle can be performed by introducing the Jacobian $\partial(x, y)/\partial(r, s)$ into the integral. The Jacobian can be approximated as a polynomial function and this may cause the order of the integrand to increase. To map the standard triangle from (ζ, s) space to an arbitrary triangle in (x, y) we will use an isoparametric mapping:

$$\begin{aligned} x(\zeta, s) &= x^{\text{vert}-a} g^1 g^2 g^3 \\ &+ x^{\text{vert}-b} g^1 g^2 g^3 \\ &+ x^{\text{vert}-c} g^1 g^2 g^3 \\ &+ \sum_n \hat{x}^1 g_n^1 g_n^2 g_n^3 \\ &+ \sum_n \hat{x}^2 g_n^1 g_n^2 g_n^3 \\ &+ \sum_n \hat{x}^3 g_n^1 g_n^2 g_n^3 \end{aligned} \quad (4)$$

We have used $\hat{x}$ to denote the expansion coefficient of the transformed coordinate along the edge. The transformation procedure is described in the next section.

Here $x^{\text{side}}, y^{\text{side}}$ and $x^{\text{vert}}, y^{\text{vert}}$ refer to the value of x, y values along the sides and vertices as given in Fig. 3. For straight sided triangles in (x, y) space we find that the Jacobian is given by:

$$\begin{aligned} J \equiv \frac{\partial(x, y)}{\partial(r, s)} &= \left[\frac{x^{\text{vert}-b} - x^{\text{vert}-a}}{2} \right] \left[\frac{y^{\text{vert}-c} - y^{\text{vert}-a}}{2} \right] \\ &- \left[\frac{x^{\text{vert}-c} - x^{\text{vert}-a}}{2} \right] \left[\frac{y^{\text{vert}-b} - y^{\text{vert}-a}}{2} \right] \end{aligned} \quad (5)$$

which is a constant. It can be appreciated that the geometric factors $\partial x/\partial r, \partial x/\partial s, \partial y/\partial r, \partial y/\partial s$ are also constant for linear triangles which proves beneficial when evaluating derivative terms (see Section 3.1.1).

For curved triangular elements neither the Jacobian nor the geometric factors are constant, which destroys some of the banded structure of the mass matrix. However, curved elements are typically required only at the boundaries and it is possible to use linear elements to discretize the interior of the solution domain. Therefore, the number of curved elements is generally sufficiently small to ensure that the increase in work for curved elements is acceptable. We note that in this case the Jacobian is approximated by a polynomial expansion which raises the order of the integrand. Since the number of quadrature points is arbitrary we could increase the integration order to compensate for the introduction of the polynomial Jacobian in these elements. Nevertheless, the error incurred by evaluating the integrands at the same quadrature points is consistent with the approximation error for smooth solutions [22]. This has been observed in our numerical experiments and thus we do not incur any “quadrature crimes” [23].

2.2.2. Notation and transformations

(a) *Local operations.* In order to describe the following algorithms we wish to define some notation in the form of vectors and matrices which perform various operations in a discrete fashion. While the exact numerical operations may not be performed as matrix vector operations, since many of the matrices will be sparse or even diagonal, it is easier to interpret the algorithms in this form than by using indice notation. Specific details on numerical implementation have been detailed in [24]. We will use the notation $\mathbf{f}$ to represent a vector of function values of $f(\zeta, s)$ at the quadrature points given by ζ_i and s_j , where by choice the i indice runs fastest. This vector is therefore $q_\zeta \cdot q_s$ long where q_ζ is the number of quadrature points in the ζ direction and q_s is the number of quadrature points in the s direction. The vector can be interpreted as the collocation evaluation of the function at quadrature points. We will use the notation $\hat{\mathbf{f}}$ to represent a vector of the expansion coefficients $\hat{f}_{mn}$. This vector is of size $M(M+1)/2$ where $(M-1)$ is the maximum order of the expansion polynomial along an edge. The storage convention for $\hat{f}_{mn}$ is to store vertex a then side 1, followed by vertex b and side 2, and similarly vertex c and side 3. Finally, the interior modes are stored with the index n running fastest. The only particular requirement for this storage convention is that the interior modes need to be stored in this way for the sparsity of the mass and stiffness matrices to be evident.

We now introduce two matrices W and G . W is a diagonal matrix containing the quadrature weights required to integrate $\mathbf{f}$ over the T^2 space. G is a matrix whose columns are the discrete values of each expansion polynomial at the quadrature points. So the matrices have the form:

$$W = \begin{bmatrix} w_0^{0,0} w_0^{1,0} & 0 & \cdots & 0 & 0 \\ 0 & w_1^{0,0} w_0^{1,0} & \cdots & 0 & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & \cdots & w_{q_\zeta-2}^{0,0} w_{q_s-1}^{1,0} & 0 \\ 0 & 0 & \cdots & 0 & w_{q_\zeta-1}^{0,0} w_{q_s-1}^{1,0} \end{bmatrix},$$

$$G = \begin{bmatrix} g_{1,0}^{1-2}(\zeta_0, s_0) & \cdots & g_{1,n}^{1-2}(\zeta_0, s_0) & \cdots & g_{mn}^{1-2}(\zeta_0, s_0) \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ g_{1,0}^{1-2}(\zeta_{q_\zeta-1}, s_0) & \cdots & g_{1,n}^{1-2}(\zeta_{q_\zeta-1}, s_0) & \cdots & g_{mn}^{1-2}(\zeta_{q_\zeta-1}, s_0) \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ g_{1,0}^{1-2}(\zeta_{q_\zeta-1}, s_{q_s-1}) & \cdots & g_{1,n}^{1-2}(\zeta_{q_\zeta-1}, s_{q_s-1}) & \cdots & g_{mn}^{1-2}(\zeta_{q_\zeta-1}, s_{q_s-1}) \end{bmatrix}.$$

If we let $N_q = q_\zeta \cdot q_s$ and $M_T = M(M+1)/2$ then W is a N_q by N_q matrix and G is a N_q by M_T matrix. Having defined $\mathbf{u}, \hat{\mathbf{u}}, W, G$ we are now in a position to form the discrete inner product in matrix notation:

$$(u(r, s), v(r, s))_{T^2} = (W\mathbf{u})^t \mathbf{v} = \mathbf{u}^t W^t \mathbf{v} = \mathbf{u}^t W \mathbf{v}.$$

The last step has used the fact that W is diagonal and therefore symmetric. This operation is exact if $u(r, s), v(r, s) \in G_M$ (i.e. within the polynomial space spanned by g_{mn}^{1-2}) and $q_\zeta = q_s = M+1$ assuming

Gauss–Lobatto integration. The inner product of the expansion mode g_{mn}^{1-2} with a polynomial function $u(r, s)$ is given by:

$$(g_{mn}^{1-2}, u(r, s))_{T^2} = (g_{mn}^{1-2})^t W u$$

and so the vector of inner products of all modes with respect to the function $u(r, s)$ is given by:

$$(g_{mn}^{1-2}, u(r, s))_{T^2} = G^t W u.$$

We are now in a position to introduce the forward transformation from *physical space* to *transformed space* by which we mean expressing a polynomial function $u(r, s)$ in terms of the expansion basis g_{mn}^{1-2} . Given a function $u(r, s)$, which lies in polynomial space G_M , we know that it can be expressed in terms of the expansion basis i.e:

$$u(r, s) = \sum_{mn} \hat{u}_{mn} g_{mn}^{1-2}(r, s). \quad (6)$$

To determine the expansion coefficients we can take the inner product of both sides of Eq. (6) with each expansion mode. In matrix notation this is expressed as :

$$G^t W u = G^t W G \hat{u}.$$

Therefore, the forward transformation is given by:

$$\hat{u} = (G^t W G)^{-1} G^t W u. \quad (7)$$

Eq. (6) clearly defines the transformation from transformed space to physical space. The backward transformation from transformed space to the quadrature points in physical space can be written in matrix notation as:

$$u = G \hat{u}.$$

It is interesting to note that for standard quadrilateral spectral elements a *nodal* basis is used and thus the expansion coefficients are simply the function values at the quadrature points [10]. In particular the expansion basis is made up of products of Lagrange polynomials at the zeros of the Gauss–Lobatto–Legendre quadrature points. Also, the inner product is evaluated using quadrature weights at the same points. The quadrature is not exact for the inner product of two modes but the quadrature error is of the same order as the approximation error for smooth functions [25]. For quadrilateral spectral elements the G matrix turns into the identity matrix, due to the dirac delta nature of the Lagrange polynomials, and the forward transformation is:

$$\hat{u} = (I^t W I)^{-1} I^t W u = W^{-1} W u = u.$$

If the function $u(r, s)$ is not contained in the polynomial space G_M then the forward and backward transformation define the projection of the function onto the expansion space. So we can consider the projection of a function onto the expansion space and subsequent evaluation at the quadrature points as the matrix operation:

$$u_M(r, s) = P u(r, s) = G \hat{u} = G (G^t W G)^{-1} G^t W u.$$

This projection clearly has the property $PPu = Pu$. Dubiner [9] describes this as a collocation projection (which is used in evaluating the values at quadrature points) followed by a Galerkin projection. So it is a Galerkin method in the degenerate inner product:

$$(u, v)_{N_q} = (P_c u, P_c v)$$

where N_q is the total number of collocation points and P_c is collocation projection which in this case evaluated the value of the function at the quadrature points. A pure collocation projection is difficult to implement since there are more quadrature points than expansion coefficients. Dubiner also interprets this projection as a spectral method with a collocation projection but where the extra-high frequencies have been filtered by a Galerkin projection.

(b) *Global operations.* When projecting a domain made up of multiple triangular subdomains a similar operation is performed. However, there are two major distinctions; the first is that the local triangular subdomains are mapped onto the standard triangle and this operation introduces a Jacobian into the integrand of the inner product. To denote the introduction of the Jacobian we use the matrix B instead of W where B is a diagonal matrix whose components are the product of the Jacobian at the quadrature points multiplied by the weight matrix, i.e. $B = JW$. As noted in Section 2.2.1, for linear subdomains the Jacobian is a constant and therefore the local forward transformation is unchanged. This is easily illustrated if we assume that $J = \alpha I$, where α is a scalar quantity, then the transformation becomes:

$$\hat{u} = (G^t B G)^{-1} G^t B u \quad \text{and} \quad B = JW = \alpha IW = \alpha W$$

so

$$\hat{u} = (G^t \alpha W G)^{-1} G^t \alpha W u = (G^t W G)^{-1} G^t W u.$$

The second distinction does change the forward transformation. Since we require the expansion basis to be C^0 continuous over the complete domain we have to ensure that the expansion coefficients of adjoining elements have the same magnitude. This operation reduces the degrees of freedom of the system. Initially, we start with KM_T degrees of freedom, where K is the number of triangular elements and M_T is the total number of expansion modes in each element. This is reduced to $KM_T - EN_{\text{edge}} - \sum_{i=0}^{\text{vertices}} (V_i - 1)$ degrees of freedom, where E is the number of connecting edges, N_{edge} is the number of modes along an edge and V_i is the multiplicity of the i^{th} vertex. If we let $\hat{u}_1$ denote all the local expansion coefficients over K elements and $\hat{u}_g$ denote the global expansion coefficients, then we can represent the projection of global degrees of freedom to local ones by the matrix operation Z :

$$\hat{u}_1 = Z \hat{u}_g.$$

Z is a non-square very sparse matrix whose values are either 1 or -1 depending on the shape of connecting modes. We recall that the odd modes of some edge configurations can be the negative of each other (see Section 2.1.2). We also note that the absolute column sum of Z is equal to the multiplicity of the respective global degree of freedom. Now let us define:

$$A = \begin{bmatrix} (G^t B G)_1 & 0 & \cdots & 0 \\ 0 & (G^t B G)_2 & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & \cdots & (G^t B G)_K \end{bmatrix} \quad (8)$$

$$B = \begin{bmatrix} (G^t B)_1 & 0 & \cdots & 0 \\ 0 & (G^t B)_2 & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & \cdots & (G^t B)_K \end{bmatrix}.$$

The subscript on each block diagonal component denotes the local matrices from each of the K elements. Therefore, the local transformation for K elements is the solution of the equation:

$$A \hat{u}_1 = B u_1. \quad (9)$$

Here u_1 denotes a vector of the function $u(r, s)$ at the quadrature points over all K elements. We wish to solve this system for the global expansion coefficients in order to impose the C^0 continuity condition and so Eq. (9) can be written:

$$AZ \hat{u}_g = B u_1. \quad (10)$$

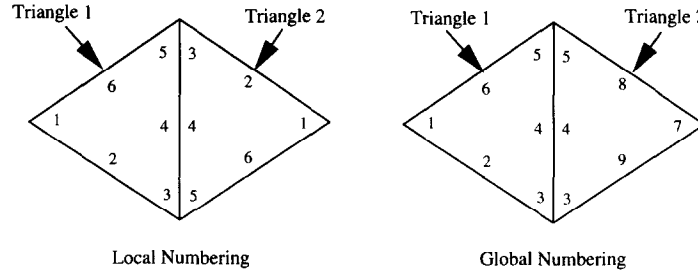


Fig. 6. Illustration of local and global numbering for a domain containing two triangular elements. Here the expansion order is $M = 3$ so there are $3(M - 1) + (M - 3)(M - 2)/2 = 6$ modes in each element, three vertex modes (1, 3, 5) and three edge modes (2, 4, 6).

The matrix AZ of course is not square, having dimensions $N_{\text{global}} \cdot N_{\text{local}}$ where N_{global} is the number of global modes and N_{local} is the local number of modes over the K elements. However, if we premultiply Eq. (10) by Z^t we obtain the square system:

$$Z^t AZ \hat{u}_g = Z^t B u_1.$$

So the global forward transformation is given by:

$$\hat{u}_g = (Z^t AZ)^{-1} Z^t B u_1. \quad (11)$$

The effect of premultiplying Eq. (10) by Z^t is to add or subtract the rows which describe the same mode on equation (9) together.

To illustrate the form of the global to local projection matrix Z we can consider the example shown in Fig. 6. We consider a global domain made up of two triangular elements. The expansion order is equal to $M = 3$ and so the total number of modes in each element is $M_T = M(M + 1)/2 = 6$. There are three vertex modes (1, 3, 5) and three edge modes (2, 4, 6) but no interior mode (7). For this case Z is an 9 by 12 matrix as shown below:

$$\hat{u}_1 = \begin{bmatrix} \hat{u}_1^1 \\ \hat{u}_1^2 \\ \hat{u}_1^3 \\ \hat{u}_1^4 \\ \hat{u}_1^5 \\ \hat{u}_1^6 \\ \dots \\ \hat{u}_2^1 \\ \hat{u}_2^2 \\ \hat{u}_2^3 \\ \hat{u}_2^4 \\ \hat{u}_2^5 \\ \hat{u}_2^6 \end{bmatrix} = \begin{bmatrix} 1 & & & & & & & & & & & \\ & 1 & & & & & & & & & & \\ & & 1 & & & & & & & & & \\ & & & 1 & & & & & & & & \\ & & & & 1 & & & & & & & \\ & & & & & 1 & & & & & & \\ & & & & & & 1 & & & & & \\ \dots & & & & & & & \dots & & & & \\ & & & & & & & & 1 & & & \\ & & & & & & & & & 1 & & \\ & & & & & & & & & & 1 & \\ & & & & & & & & & & & 1 \end{bmatrix} \begin{bmatrix} \hat{u}_g^1 \\ \hat{u}_g^2 \\ \hat{u}_g^3 \\ \hat{u}_g^4 \\ \hat{u}_g^5 \\ \hat{u}_g^6 \\ \hat{u}_g^7 \\ \hat{u}_g^8 \\ \hat{u}_g^9 \end{bmatrix}.$$

The superscripts denote the local or global nodal number and the subscript denote the element number. As stated previously the absolute column sum gives the multiplicity of a mode and we see that as expected columns 3, 4 and 5 all have a multiplicity of 2. We also note that the absolute row sum is always 1 since there is only ever one value of each local mode. If there had been some odd edge modes being matched there would have been a possibility of a (-1) entry. In general, a projection using the local forward transformation given by Eq. (7) will not be the same as a projection using the global forward transformation as defined by equation (11) since the local transformation does not guarantee a global C^0 expansion. However, if the function

$u(r, s)$ lies in the expansion space G_M the transformations are identical because the function can be completely expressed by the expansion basis and so the C^0 condition is implicitly satisfied.

As a final point we note that we can use this notation to show the global mass matrix $Z^t A Z$ is positive definite, i.e.

$$v^t (Z^t A Z) v > 0 \quad \forall \text{ nonzero vectors } v.$$

To show this we consider the Legendre inner product of a non-zero function $u(x, y)$ which exists in the C^0 global expansion space. Since the function is non-zero we know that:

$$\int_{\Omega} u(x, y) u(x, y) = \sum_{k=1}^K \int_{T_k^2} u(r^k, s^k) u(r^k, s^k) > 0.$$

We can exactly represent the integration in a discrete form by introducing the local vectors u_1^k which contain the evaluation of the function $u(r^k, s^k)$ at the quadrature points. So we have:

$$\sum_{k=1}^K \int_{T_k^2} u(r^k, s^k) u(r^k, s^k) = \sum_{k=1}^K (u_1^k)^t B^k u_1^k > 0.$$

Now since the function exist in the expansion space we can represent u_1^k and $G u_1^k$ and we therefore have:

$$\sum_{k=1}^K (u_1^k)^t B^k u_1^k = \sum_{k=1}^K (\hat{u}_1^k)^t (G^t B G)^k \hat{u}_1^k = (\hat{u}_1)^t A \hat{u}_1 > 0.$$

Finally, we use fact that $\hat{u}_g = Z \hat{u}_1$ to arrive at the desired result:

$$(\hat{u}_g)^t (Z^t A Z) \hat{u}_g > 0.$$

Since only the constraints put on $u(x, y)$ were that it was non-zero and existed in the global expansion space, any function $u(x, y)$ can be described by the global expansion coefficients $\hat{u}_g$ multiplied by the expansion space $1-2$
 g_{mn} . Therefore, the above condition is true for all non-zero vector $\hat{u}_g$ and so the global mass matrix $Z^t A Z$ is positive definite.

2.2.3. Differentiation

Differentiation can be performed in both the transformed and physical space. In physical space when a function is evaluated at the quadrature points Dubiner [9] points out that the function belongs to the space G_Q defined as:

$$G_Q = \text{Span}\{r^m s^n\}_{(mn) \in Q}$$

where $Q = \{(m, n) | 0 \leq m \leq q_r; 0 \leq n \leq q_s\}$, $q_r \geq M$, $q_s \geq N$.

Differentiation in G_Q can be evaluated using the derivatives of the Lagrange polynomials at the quadrature points, as is typically done in quadrilateral spectral elements. Also since $G_M \subset G_Q$ evaluation of the derivatives in this fashion will be complete for any projected function. However, the quadrature points are placed at points of constant ζ and s rather than r and s so to evaluate the local r and s derivatives we apply the chain rule. The local gradient operator is therefore:

$$\nabla = \begin{pmatrix} \frac{\partial}{\partial r} \\ \frac{\partial}{\partial s} \end{pmatrix} = \begin{pmatrix} \frac{\partial \zeta}{\partial r} \frac{\partial}{\partial \zeta} \Big|_s \\ \frac{\partial \zeta}{\partial s} \frac{\partial}{\partial \zeta} \Big|_s + \frac{\partial s}{\partial s} \frac{\partial}{\partial s} \Big|_{\zeta} \end{pmatrix} = \begin{pmatrix} \frac{2}{(1-s)} \frac{\partial}{\partial \zeta} \Big|_s \\ \frac{(1+\zeta)}{(1-s)} \frac{\partial}{\partial \zeta} \Big|_s + \frac{\partial}{\partial s} \Big|_{\zeta} \end{pmatrix}$$

where $\zeta = 2(1+r)/(1-s) - 1$.

This allows us to calculate the derivatives at the quadrature points of any function in the expansion space G_M except at the singular point ($r = -1, s = 1$). Here the derivative's co-efficients $2/(1-s)$, $(1+\zeta)/(1-s)$ are infinite. If we choose the Gauss points to evaluate the inner product integral this would not be a problem as we would not have a quadrature point at the singularity. However, if we use the Gauss–Lobatto points, which makes it easier to impose boundary conditions, then we need to evaluate this derivative. This is easily achieved if we evaluate this derivative in the transformed space since the $1/(1-s)$ factor can be factored out explicitly. In transformed space the derivative may be evaluated as the summation of the expansion coefficients multiplied by the derivative of the expansion modes, i.e.

$$\frac{\partial u(r, s)}{\partial r} = \sum_{mn} \hat{u}_{mn} \frac{2}{(1-s)} \left. \frac{\partial}{\partial \zeta} \frac{1-\zeta^2}{8}(\zeta, s) \right|_{mn}.$$

Whilst at first this may look expensive to evaluate, it turns out that this is not the case since most derivatives of the modes vanish at ($r = -1, s = 1$). As can be seen in Section 2.1.2, the interior modes and all the edge modes on side 1 include the factor $(\frac{1-s}{2})^m$, $m \geq 2$. Thus, both the $\partial/\partial r$ and $\partial/\partial s$ derivatives are zero for this set of modes at $s = 1$. Therefore, the derivatives can be expressed as the following summations over the coefficients of the vertices and edges 2 and 3 (see Fig. 3):

$$\begin{aligned} \left. \frac{\partial u(r, s)}{\partial r} \right|_{(-1,1)} &= \frac{1}{2} \left[\hat{u}^{\text{vert}-b} - \hat{u}^{\text{vert}-a} + \sum_{n=1}^{M-1} P_{n-1}^{1,1}(1) (\hat{u}_{1,n}^{\text{side}-2} - \hat{u}_{1,n}^{\text{side}-3}) \right] \\ \left. \frac{\partial u(r, s)}{\partial s} \right|_{(-1,1)} &= \frac{1}{2} \left[\hat{u}^{\text{vert}-c} - \hat{u}^{\text{vert}-a} - \sum_{n=1}^{M-1} P_{n-1}^{1,1} \hat{u}_{1,n}^{\text{side}-3} \right] \end{aligned}$$

This reduces the operation count to evaluate the derivative at the singular corner to $O(M)$ provided that, as is normally the case, we know the expansion coefficients. If we do not know the expansion coefficients then, providing the function is in polynomial space spanned by the expansion basis (i.e in G_M), we can simply perform a one-dimensional transformation along sides 2 and 3 which involves an order $O(M^2)$ operation.

Finally, we can compare the evaluation of the derivative $\partial u/\partial r$ in physical space for quadrilateral and triangular spectral elements. Consider a function represented over a bi-unit square by the basis:

$$u(r, s) = \sum_i \sum_j u_{i,j} h_i(r) h_j(s)$$

as is the case in quadrilateral spectral elements [10]. Here $h_i(z)$ represents the Lagrange polynomial through the i^{th} Gauss–Lobatto–Legendre quadrature point. Now the r partial derivative at the k, l point quadrature (keeping s constant) is given by:

$$\left. \frac{\partial u(r_k, s_l)}{\partial r} \right|_{s_l} = \sum_i \sum_j u_{i,j} \frac{\partial h_i(r_k)}{\partial r} h_j(s_l) = \sum_i u_{i,l} \frac{\partial h_i(r_k)}{\partial r}.$$

Here we have used the property $h_i(z_j) = \delta_j^i$ where δ_j^i is the Kronecker delta. If we define a matrix D with entries:

$$D_{ij} = \frac{\partial h_j(r_i)}{\partial r},$$

then the partial derivative of $u(r, s)$ with respect to r at all quadrature points can be expressed as a matrix operation on $u(r, s)$ at the quadrature points by:

$$u_r = D_r u$$

where

$$D_r = \begin{bmatrix} D & 0 & \dots & 0 \\ 0 & D & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & D \end{bmatrix},$$

u_r, u being the value of $\partial u / \partial r, u$ at the quadrature points, where the quadrature points run fastest in the r direction. Now if we consider the triangular spectral element case, the expansion in physical space can be expressed as:

$$u(r, s) = \sum_i \sum_j u_{i,j} h_i(\zeta) h_j^{1,0}(s),$$

where $h_j^{1,0}(s)$ is the j^{th} Lagrange interpolant through the zeros of the $(1, 0)$ Jacobi polynomial and the point $s = \pm 1$. The partial derivative with respect to r at the k, l quadrature point keeping s constant is:

$$\left. \frac{\partial u(r_k, s_l)}{\partial r} \right|_{s_l} = \sum_i \sum_j u_{i,j} \frac{2}{(1-s_l)} \frac{\partial h_i(\zeta_k)}{\partial \zeta} h_j(s_l) = \sum_i u_{i,l} \frac{2}{(1-s_l)} \frac{\partial h_i(\zeta_k)}{\partial \zeta}.$$

So we can express this as the matrix operation:

$$u_r = \bar{D}_r u$$

where

$$\bar{D}_r = \begin{bmatrix} c_0 D & 0 & \dots & 0 \\ 0 & c_1 D & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & c_{q_s-1} D \end{bmatrix},$$

$$c_i = \frac{2}{(1-s_i)}.$$

If the eigenvalues of the matrix D are denoted by the vector λ then the eigenvalues of D_r are also given by λ but each eigenvalue has a multiplicity equal to the number of s quadrature points since the D_r matrix is made up of q_s decoupled D matrices. However, the eigenvalues of the matrix $\bar{D}_r$ are given by $c_0 \lambda, c_1 \lambda, \dots, c_{q_s-1} \lambda$. Canuto *et al.* [25] scale the modulus of the first order differential matrix with homogeneous Dirichlet boundary conditions by:

$$|\lambda| = O(q^2)$$

where q is equal to the number of quadrature points in the r direction. Therefore, the two-dimensional quadrilateral derivative matrix D_r will have a similar scaling. However, the triangular derivative matrix $\bar{D}_r$ has an extra c_i coefficient. It can be shown that the distance of the nearest quadrature point from the point where $s = 1$ is of the order $O(1/q_s^2)$, where q_s is the total number of quadrature points. So the coefficient c_i has a maximum order given by:

$$|c_i| \approx \left| \frac{2}{(1-s_i)} \right| \approx O(q_s^2).$$

Therefore, the scaling of the eigenvalues of $\bar{D}_r$ is $O(q_s^2 q^2)$. Typically $q_s \approx q$ and thus the scaling becomes $O(q^4)$. This is undesirable for time dependent equations involving a convective operator since explicit treatment of the convective operator forces the time step to scale with the reciprocal of the largest eigenvalue. Nevertheless, in the transformed space the convective operator is scaled by the basis when the inner product is taken. This has the effect of recovering the $O(q^2)$ scaling of the spectrum.

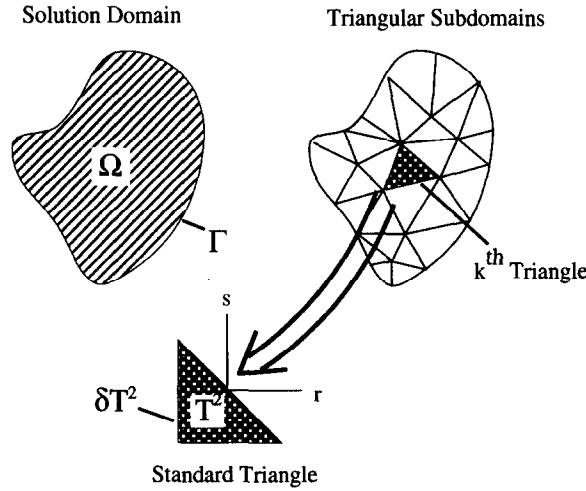


Fig. 7. The solution domain is divided into triangular subdomains which are then mapped to the standard triangle as defined in Fig. 2. All operations are then performed in the local (r, s) space on the standard triangle.

3. Advection and Helmholtz equations

In this section we apply the previous formulation to the linear advection and Helmholtz equations. The advection equation allows us to identify the time step restrictions which arise from the hyperbolic nature of this equation when dealing with explicit evaluation of the convective terms. To determine this restriction we will construct the eigenspectrum of the weak derivative operators. In addition, the advection equation allows us to evaluate the dissipative properties of the discrete operator. We will also consider the elliptic Helmholtz equation since this is a prototype to all elliptic contributions to the Navier–Stokes equations.

3.1. Linear advection equation

We want to consider the two-dimensional linear advection equation for $u(x, y; t)$:

$$\frac{\partial u}{\partial t} + Lu = \frac{\partial u}{\partial t} + (\mathbf{V} \cdot \nabla)u = 0 \quad (12)$$

written in full as :

$$\frac{\partial u(x, y; t)}{\partial t} + a(x, y) \frac{\partial u(x, y; t)}{\partial x} + b(x, y) \frac{\partial u(x, y; t)}{\partial y} = 0,$$

$$\mathbf{V} = [a(x, y), b(x, y)], \quad u(x, y; 0) = u_0(x, y).$$

The terms $a(x, y)$, $b(x, y)$, $u_0(x, y)$ are real, u_0 is considered to be smooth and a, b will usually be considered to be constants and divergence free (i.e. $\nabla \cdot \mathbf{V} = 0$). The equation is also supplemented with appropriate boundary conditions which will either be periodic or Dirichlet inflow.

3.1.1. Discretization

As shown in Fig. 7 we will consider the solution on a domain, denoted by Ω , which is fixed in space for all time and has a boundary Γ . Initially we divide the domain into K triangular subdomains denoted by T_k^2 each of which has a boundary ∂T_k^2 . Therefore, the union of the K subdomains T_k^2 is equal to Ω i.e:

$$\Omega = \bigcup_k^K T_k^2$$

and the domain boundary, Γ , is contained within the union of all boundaries, ∂T_k^2 , i.e:

$$\Gamma \subset \bigcup_k^K \partial T_k^2.$$

If we let the local co-ordinates of the k^{th} subdomain be (r^k, s^k) then T_k^2 has already been defined as

$$T_k^2 = \{(x, y) | -1 \leq r^k, s^k; r^k + s^k \leq 0\},$$

where (x, y) is related to (r^k, s^k) by a one-to-one linear mapping given in (4).

The global expansion basis $g_{mn}^{1-2}(x, y)$ can be defined in terms of the local expansion functions $g_{mn}^{1-2}(r^k, s^k)$ in the following manner: since the interior functions are zero at the boundaries and are therefore already C^0 continuous we can define $g_{mn}^{1-2}(x, y)$ for interior (m, n) (i.e. $2 \leq m, 1 \leq n, m < M, n + m < M$) as

$$g_{mn}^{1-2}(x, y) = \begin{cases} g_{mn}^{1-2 \text{ interior}}(r^k, s^k) & , \quad (x, y) \in T_k^2; \\ 0 & \text{otherwise} \end{cases} \quad (m, n) \text{ interior.}$$

The side and vertex modes are not so readily transcribed but are nonetheless easily interpreted. Along an edge where two subdomains k and l meet (i.e. $\partial T_k^2 \cap \partial T_l^2 \neq \emptyset$), if side i meets a side j then the global mode is given by:

$$g_{mn}^{1-2k}(x, y) = g_l^{1-2 \text{ side-}i}(r^k, s^k) \pm g_l^{1-2 \text{ side-}j}(r^l, s^l)$$

where l is either m or n depending on which sides are connected and the sign is chosen to make the mode continuous. The global vertex modes are evaluated in a similar fashion except there is a summation of local vertex modes equal to the multiplicity of the vertex. There is no sign ambiguity involved in the vertex summation.

The relationship between the global and local expansion coefficients was also addressed in the previous section. The projection from global to local expansion coefficients was defined in terms of matrix operator Z . The introduction of this operator allowed us to determine the global matrix system for the forward transformation by constructing the local systems and then operating on the local matrices by pre- and post-multiplying by Z^t and Z , respectively. This gave us a global forward transformation matrix for a C^0 continuous global expansion.

The entire operation can be interpreted in two similar ways. The first is that a local operation is being performed on each element or subdomain which is constrained by the condition that the global solution be C^0 continuous. This constraint is satisfied by matching the expansion coefficients of the modes that meet along adjacent elemental boundaries. The second interpretation is simply that the transformation is being performed with the global C^0 continuous basis as described above. These two interpretations may lead to the same discrete system, although the first interpretation allows a simpler practical implementation of numerical algorithms. However, the second interpretation helps to explain the motivation behind some of the operations involved in constructing the algorithm. For example, the motivation for pre- and post-multiplying the local forward transformation system by Z^t and Z was to add/subtract the equation and modes that needed to be matched. This operation could be achieved by averaging in any variety of fashions, however the use of Z can also be interpreted as constructing the inner product between the respective global C^0 boundary modes.

Now if we consider only the k^{th} triangular subdomain (dropping any k index at present) locally our solution is approximated by:

$$u_M(x, y; t) = Pu(x, y; t) = \sum_{mn}^M \hat{u}(t) g_{mn}^{1-2}(r(x, y), s(x, y)),$$

where the projection P was introduced in Section 2.2.2 and we have used M to denote a finite truncation of the expansion thereby implying an approximation. Also, we recall that the projected solution u_M can be represented, at a given time t , in transformed space as the product of the expansion coefficients $\hat{u}(t)$ and the local expansion basis $g_{mn}^{1-2}(r, s)$. Substituting u_M into Eq. (12) we have:

$$\frac{\partial u_M}{\partial t} + Lu_M = 0.$$

In general, we cannot guarantee that $\partial u_M / \partial t$ will remain in the global expansion space. For example, the derivative of the global expansion basis is discontinuous at the elemental boundaries and so it does not belong in the C^0 global expansion space. Therefore, we take the inner product of the equation with the expansion basis and we have the system of equations:

$$\left(g_{mn}^{1-2}(r, s), \frac{\partial u_M}{\partial t} \right) + \left(g_{mn}^{1-2}(r, s), Lu_M \right) = 0 \quad \forall(m, n).$$

We can write this system of equations in matrix form:

$$G^t W u_t + G^t W L u = 0,$$

where u_t and u are vectors of $\frac{\partial}{\partial t} u_M(r, s; t)$ and $u_M(r, s; t)$ at the quadrature points, respectively. Since

$$u_M = \sum_{mn} \hat{u}(t) g_{mn}^{1-2}(r, s)$$

which has the matrix form:

$$u_M(r, s; t) = G(r, s) \hat{u}(t).$$

We can then form the Galerkin approximation to (12) in terms of the expansion coefficients as:

$$G^t W G \hat{u}_t + G^t W L G \hat{u} = 0. \quad (13)$$

At this point we need to define the discrete version of the operator $L \equiv A \cdot \nabla$. In the above equation we see that L operates on the expansion basis G . Since G is a polynomial, differentiation can be performed exactly by differentiating the Lagrange polynomial through the quadrature points as explained in Section 2.2.3. We can represent the differentiation with respect to r and s at the quadrature points by the matrix operations:

$$u_r = D_r u, \quad u_s = D_s u$$

where u_r and u_s are vectors of $\frac{\partial}{\partial r} u_M(r, s; t)$ and $\frac{\partial}{\partial s} u_M(r, s; t)$, respectively. The form of the matrix D_r is described in Section 2.2.3 and the matrix D_s is constructed in a similar fashion. The operator written in full is:

$$L \equiv a(x, y) \frac{\partial}{\partial x} + b(x, y) \frac{\partial}{\partial y}.$$

If we use the chain rule on the x, y derivatives it can be expressed in terms of r, s derivatives as:

$$L = \left(a(x, y) \cdot \frac{\partial r}{\partial x} + b(x, y) \cdot \frac{\partial r}{\partial y} \right) \frac{\partial}{\partial r} + \left(a(x, y) \cdot \frac{\partial s}{\partial x} + b(x, y) \cdot \frac{\partial s}{\partial y} \right) \frac{\partial}{\partial s}.$$

Thus the discrete version of the L operator acting at the quadrature points on the polynomial expansion G can be written:

$$L G = R D_r G + S D_s G,$$

where R is a diagonal matrix whose components are the values of $(a(x, y) \cdot \frac{\partial r}{\partial x} + b(x, y) \cdot \frac{\partial r}{\partial y})$ at the quadrature points and S is a diagonal matrix whose components are the values of $(a(x, y) \cdot \frac{\partial s}{\partial x} + b(x, y) \cdot \frac{\partial s}{\partial y})$ at the quadrature points.

It should be appreciated that the solution and not the operator L is being approximated as once the solution has been projected into the expansion space the operator L is being evaluated exactly. For linear triangles the *geometric factors* $\partial r / \partial x, \partial s / \partial x, \partial r / \partial y, \partial s / \partial y$ are constants. If $a(x, y)$ and $b(x, y)$ are also constants then R and S are simply scalars.

Eq. (13) can now be written as the system of ordinary differential equations:

$$G^t W G \frac{d\hat{\mathbf{u}}}{dt} = -G^t W (R D_r + S D_s) G \hat{\mathbf{u}}$$

If we only had one triangular domain then multiplying this equation by the inverse of $(G^t W G)$ would give us the semi-discrete form of Eq. (12) but we wish to construct the approximation to the union of triangles. To do this we redefine the global matrix $\mathbf{B}$ as

$$\mathbf{B} = \begin{bmatrix} (G^t B L G)_1 & 0 & \cdots & 0 \\ 0 & (G^t B L G)_2 & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & \cdots & (G^t B L G)_K \end{bmatrix}.$$

Since the local expansion coefficients are related to the global coefficients by the matrix $\mathbf{Z}$ and the local to global projection is completed by premultiplying the equation by $\mathbf{Z}^t$, using $\mathbf{A}$ as defined by (8) we can construct the global system:

$$\mathbf{Z}^t \mathbf{A} \mathbf{Z} \frac{d\hat{\mathbf{u}}_g}{dt} = \mathbf{Z}^t \mathbf{B} \hat{\mathbf{u}}_1 = \mathbf{Z}^t \mathbf{B} \mathbf{Z} \hat{\mathbf{u}}_g.$$

The subscript g denotes global degrees of freedom over all K elements. If we multiply by the inverse of the mass matrix $\mathbf{Z}^t \mathbf{A} \mathbf{Z}$ we have the semi-discrete system for the solution of the global expansion modes:

$$\frac{d\hat{\mathbf{u}}_g}{dt} = (\mathbf{Z}^t \mathbf{A} \mathbf{Z})^{-1} \mathbf{Z}^t \mathbf{B} \mathbf{Z} \hat{\mathbf{u}}_g. \quad (14)$$

The complete discretization then involves the solution of this system of ordinary differential equations in time. We have chosen to solve this using the Adams–Bashforth multi-step scheme (see [26]).

It is an assumption of the Galerkin projection that the global expansion basis satisfies the boundary condition on Γ . This condition is imposed by fixing the expansion coefficients of the local modes which have non-zero values along the solution boundary Γ . These values are determined from a one-dimensional transformation along the boundaries where $\partial T_k^2 \cap \Gamma \neq \emptyset$. This decomposition of modes can be interpreted as splitting the solution $u(x, y; t)$ into a boundary and homogeneous function, i.e:

$$u(x, y; t) = v(x, y; t) + w(x, y)$$

where,

$$v(\Gamma; t) = 0, \quad w(\Gamma) = u(\Gamma; t).$$

The practical significance of this is that the matrix $\mathbf{Z}^t \mathbf{A} \mathbf{Z}$ will be *condensed* to exclude the boundary modes before it is inverted. The matrix vector product $\mathbf{Z}^t \mathbf{B} \hat{\mathbf{u}}_1$ can be calculated locally and then sorted into a global numbering before being operated on by the inverse of the mass matrix.

3.1.2. Spectrum of the weak convective operator

The semi-discrete form of the weak advection equation was given in Eq. (14). We wish to see how the eigenspectrum of $(\mathbf{Z}^t \mathbf{A} \mathbf{Z})^{-1} \mathbf{Z}^t \mathbf{B} \mathbf{Z}$ grows with M because this dictates what time step restrictions will be imposed if we use an explicit time stepping scheme. In order to obtain a stable scheme we require that the eigenvalues of the matrix $(\mathbf{Z}^t \mathbf{A} \mathbf{Z})^{-1} \mathbf{Z}^t \mathbf{B} \mathbf{Z}$ lie within the stability region of the time stepping scheme.

The matrix $\mathbf{B}$ consists of components of the inner product of the global expansion basis with the convective operator acting on the expansion basis, i.e. $(\mathbf{g}_{mn}^{1-2}, L \mathbf{g}_{pq}^{1-2})$. We note that:

$$L \mathbf{g}_{pq}^{1-2} = \mathbf{V} \cdot \nabla \mathbf{g}_{pq}^{1-2} = \nabla \cdot (\mathbf{V} \mathbf{g}_{pq}^{1-2})$$

where the second relation is a consequence of $\mathbf{V}$ being a divergence free field, i.e. $\nabla \cdot \mathbf{V} = 0$. Since $\mathbf{g}_{mn}^{1-2}, \mathbf{g}_{pq}^{1-2}$ are both scalar fields we can use the vector identity $(\nabla \cdot (\phi \mathbf{V})) = \phi \nabla \cdot \mathbf{V} + \nabla \phi \cdot \mathbf{V}$ to show:

$$(\mathbf{g}_{mn}^{1-2}, L \mathbf{g}_{pq}^{1-2}) = -(L \mathbf{g}_{mn}^{1-2}, \mathbf{g}_{pq}^{1-2}) + \int_{T^2} \nabla \cdot (\mathbf{g}_{mn}^{1-2} \mathbf{V} \mathbf{g}_{pq}^{1-2}) \partial A.$$

Applying the divergence theorem to the last term we have:

$$(\mathbf{g}_{mn}^{1-2}, L \mathbf{g}_{pq}^{1-2}) = -(L \mathbf{g}_{mn}^{1-2}, \mathbf{g}_{pq}^{1-2}) + \int_{\partial T^2} \mathbf{g}_{mn}^{1-2} \mathbf{g}_{pq}^{1-2} \mathbf{V} \cdot \mathbf{n} \partial S$$

where $\mathbf{n}$ is the outward normal around a triangular element. Clearly, when either $\mathbf{g}_{mn}^{1-2}$, $\mathbf{g}_{pq}^{1-2}$ are interior modes the line integral is zero since all interior modes, by definition, are zero along the edges of each element. For all elemental boundaries, which are in the interior of the solution domain, the line integral will also be zero as we have required C^0 continuity and the outward normals are equal and opposite for adjacent elements. Therefore, when the global basis is summed via the operation of the Z matrix these boundary terms will vanish. The only remaining contribution left from the line integral is from those elements at the boundary of the solution domain where $\partial T^2 \cap \Gamma \neq 0$. If these terms also cancel then the matrix of inner products of $(\mathbf{g}_{mn}^{1-2}, \mathbf{g}_{pq}^{1-2})$ will be skew-symmetric which means that the convective operator is non-dissipative. The most common form of skew-symmetric boundary conditions are periodic conditions.

The skew-symmetry of this operator implies that the eigenvalues of $Z^t \mathbf{B} Z$ will be purely imaginary if the boundary conditions are skew-symmetric. Since the mass matrix $Z^t \mathbf{A} Z$ is positive definite (see Section 2.2.2) the eigenvalue of $(Z^t \mathbf{A} Z)^{-1} Z^t \mathbf{B} Z$ will have the signs as the eigenvalues of $Z^t \mathbf{B} Z$. This is easily shown if we consider:

$$(Z^t \mathbf{A} Z)^{-1} Z^t \mathbf{B} Z x = \lambda x,$$

letting $(Z^t \mathbf{A} Z) = R^t R$ and $y = R x$ we can re-arrange the above expression to arrive at:

$$R^{-1} [(R^{-1})^t (Z^t \mathbf{B} Z) R^{-1}] y = \lambda y.$$

The matrix $(Z^t \mathbf{B} Z)$ is being pre- and post- multiplied by $(R^{-1})^t$ and R^{-1} which is a congruence transformation and therefore the multiplied system has the same number of positive, zero and negative eigenvalues as the unmultiplied system [27]. This skew symmetric case requires the time differencing scheme to have a stability region which encompasses part of the imaginary axis. The Adams–Bashforth scheme of order greater than one fulfills this requirement [26]. We can use the periodic problem to ascertain the appropriate time step restrictions which are imposed by the explicit convective operator. The third-order Adams–Bashforth scheme has a stability region which crosses the imaginary axis at:

$$\Delta t \cdot \lambda_{\max} \simeq 0.723,$$

where $\lambda_{\max}$ is the maximum permissible eigenvalue for the system to be stable. It is now evident that the growth rate of $\lambda_{\max}$ will dictate the restriction imposed on Δt .

If we consider a periodic solution domain split into two triangles as shown in Fig. 8, where one triangle is in the standard space, we can determine the maximum eigenvalue for wave speeds V at different θ angles to the horizontal. The magnitude of the wave speed is always assumed to be one (i.e. $|V| = 1$).

We have only displayed eigenvalues for the range $-90^\circ \leq \theta \leq 90^\circ$ since a velocity with components $V = [1, 1]$ will have the same maximum eigenvalue as a velocity with components $V = [-1, -1]$. It can be appreciated that the first quadrant is similar to the third quadrant and the second quadrant is similar to the fourth. From the polar plot in Fig. 8 we see that the eigenvalue of the convective operator has a maximum value at $\theta = 45^\circ$. Two other orientations were considered where the upper triangle is rotated so that the singular vertex is pointing at the other corners and identical spectra were found in each case.

A simpler example is to consider only a single domain as shown in Fig. 9. In this example we use the standard triangle and once again propagate a wave of unit velocity at an angle θ to the horizontal. Since this domain is no longer periodic we have to impose boundary conditions. As indicated in Fig. 9 we apply homogeneous inflow boundary conditions on the horizontal and vertical boundary and treat the sloping boundary as an outflow boundary. The imposition of boundary conditions means that the convective operator $(Z^t \mathbf{A} Z)^{-1} Z^t \mathbf{B} Z$

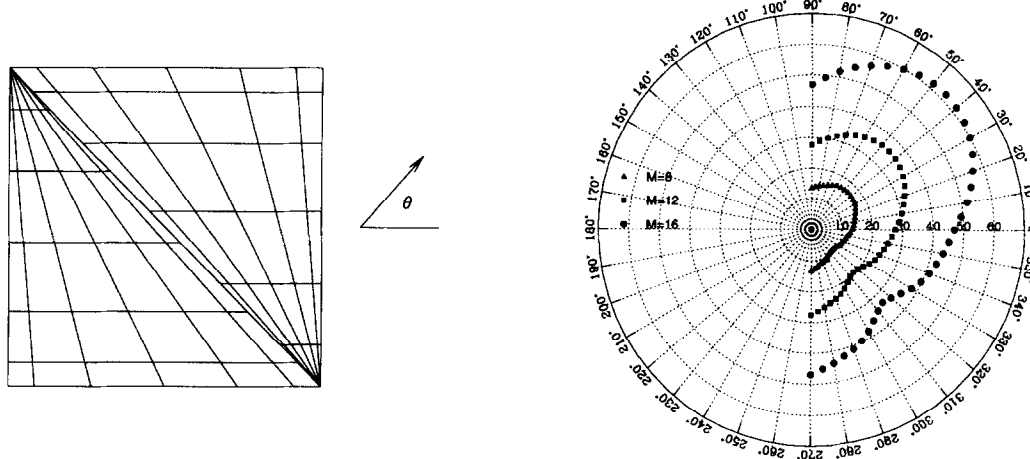


Fig. 8. For the periodic domain shown on the left (where the intersecting lines indicate the quadrature points) we consider a wave traveling at an angle θ to the horizontal with unit magnitude. For several wave orientations the maximum eigenvalue of the weak convective operator $(Z^T A Z)^{-1} Z^T B Z$ was determined and is shown on the polar plot on the right.

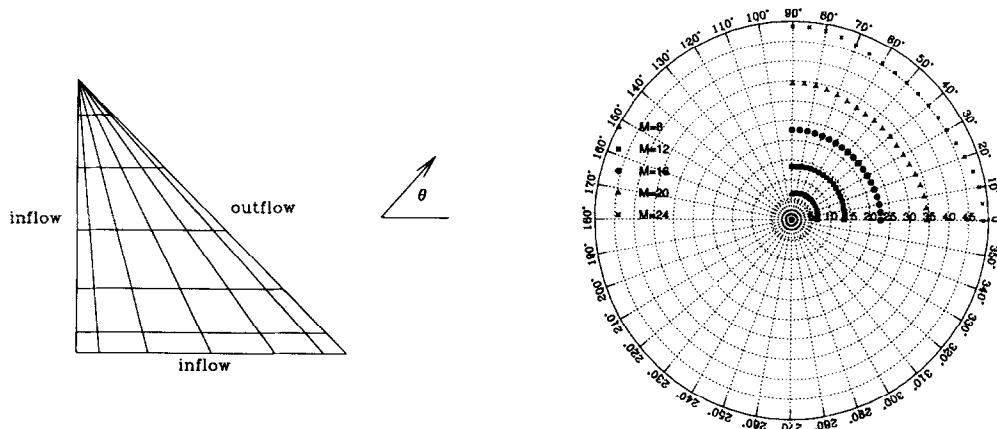


Fig. 9. For the domain shown on the left (where the intersecting lines indicate the quadrature points) we consider a wave traveling at an angle θ to the horizontal with unit magnitude. The maximum eigenvalue of the weak convective operator $(Z^T A Z)^{-1} Z^T B Z$ was determined for various wave orientations and is shown on the polar plot on the right.

is no longer skew symmetric. Nevertheless, we find that the largest eigenvalue lies on the imaginary axis (see Fig. 10) and the values of these eigenvalues for a range of wave velocities of unit magnitude at angles between $0 \leq \theta \leq 90$ are also shown in Fig. 9. The distribution of eigenvalues is far more uniform over the θ range than in the periodic box test. It is also interesting to note that for the lower expansion orders ($M = 8, 12$) the maximum eigenvalue is at $\theta = 45^\circ$ where as for the higher expansion orders the maximum eigenvalue are at $\theta = 0^\circ, 90^\circ$. This property is reflected in the spectra as shown in Fig. 10. Here we see the complete spectra for a wave of unit magnitude propagating at $\theta = 45^\circ$ to the horizontal for expansion orders of $M = 8, 16$ and 24 . The lower expansion orders ($M = 8, 12$) have eigenvalues which lie close to a semi-circle distribution in the left hand plane where as the higher expansion orders have a very different more elongated distribution.

We are interested in determining the time step restriction due to expansion order and to this aim we need a bound for the maximum eigenvalue. For both of the examples described previously the maximum eigenvalue is plotted versus expansion order in Fig. 11. This plot is on log-log axis and it is evident that for both examples the growth rate is bounded by M^2 . This leads us to the conclusion that we can bound the maximum eigenvalue by:

$$C(V, L) M^2 > \lambda_{\max},$$

where $C(V, L)$ is an expression which related the local V and characteristic length L of an arbitrary triangular

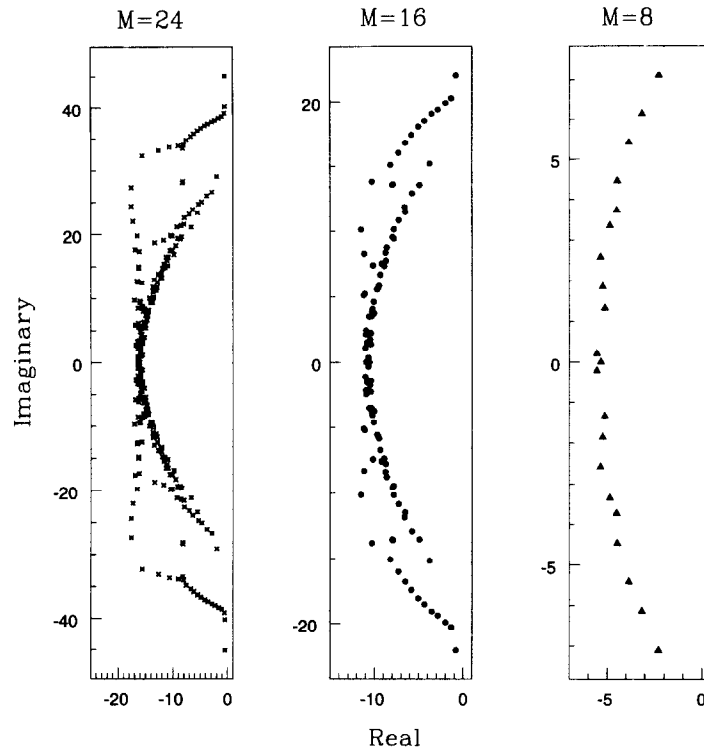


Fig. 10. The eigenspectrums of the weak convective operator $(Z^T A Z)^{-1} Z^T B Z$ for a wave of unit magnitude traveling at $\theta = 45^\circ$ at expansion orders $M = 8, 16$ and 24 .

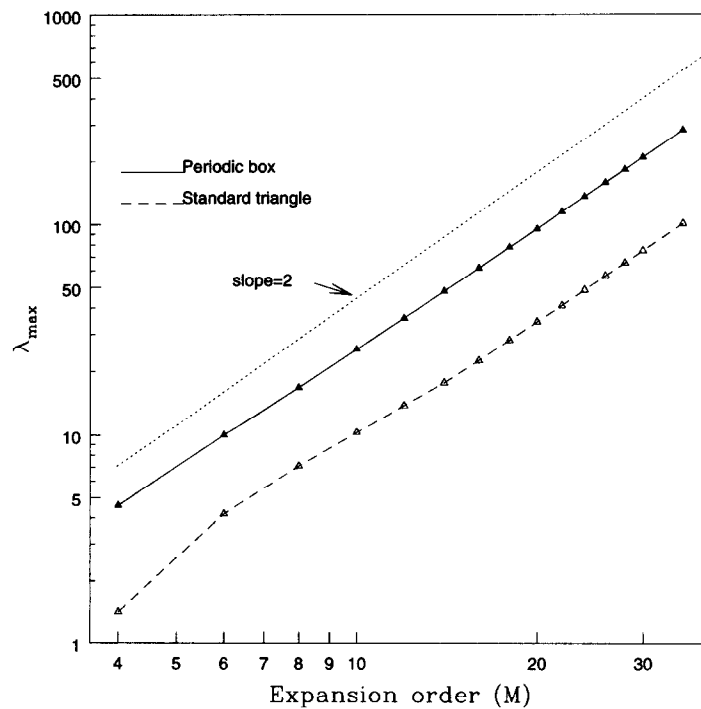


Fig. 11. Growth rate of the maximum eigenvalue $\lambda_{\max}$ with respect to expansion order M . The 'periodic box' test case is shown in Fig. 8 and the 'standard triangle' is shown in Fig. 9. In both cases the growth rate is bounded by M^2 .

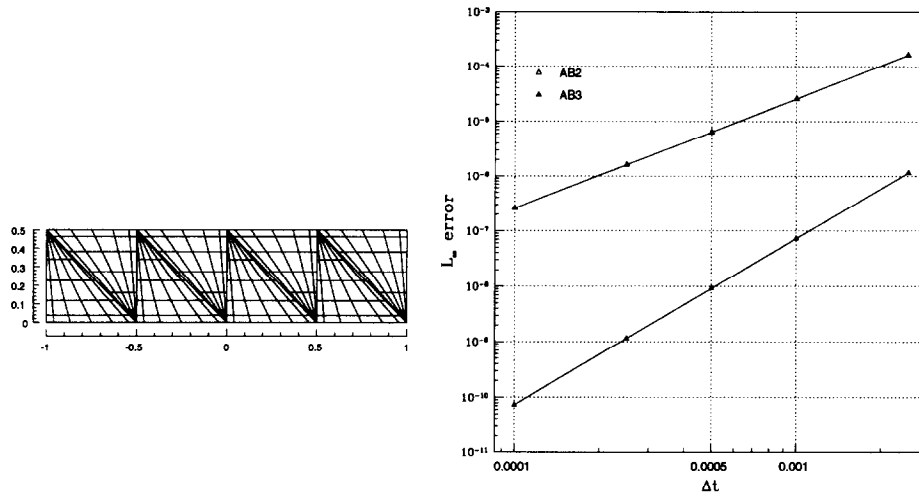


Fig. 12. Plot of the L_∞ error of the solution to the linear advection equation after one periodic cycle for the initial condition $u_0(x, y) = \sin(\pi x)$ using the mesh shown on the left. The test was performed using the Adams–Bashforth second- and third-order schemes which are verified by the slope of the lines.

domain to the standard triangle. The evaluation of $C(V, L)$ is highly problem dependent and so it is difficult to formalize an algebraic expression. Nevertheless, we now have the desired result that the time step restriction for explicit evaluation of the convective operator is bounded by:

$$\Delta t \leq 0.723 / (C(V, L)M^2)$$

when using a third order Adams–Bashforth scheme.

3.1.3. Results and validation

The first validation test we shall consider is to ensure that time accuracy is achieved. To do this we use the easily resolved initial condition

$$u_0(x, y) = \sin(\pi x)$$

on the solution domain shown in Fig. 12. Using an expansion order of $M = 12$ we can represent the initial condition in the L_∞ norm to within approximately 1×10^{-12} . Therefore, any errors higher than this can be expected to be due to time accuracy. Here we determine the L_∞ error in the solution after a periodic total time of $T = 2$ using the Adams–Bashforth second- and third-order accurate schemes. The results of this test are given in Fig. 12 for both time stepping schemes. The slope of the second-order scheme on logarithmic axes is exactly 2 and similarly the slope of the third-order scheme is exactly 3. As a final point, we note that exact information was required for the two previous time steps at start up in order to ensure third-order accuracy over the entire solution period.

The second test examines spatial convergence due to polynomial order as well as to the number of elements. We consider the initial condition

$$u_0(x, y) = \sin(\pi \cos(\pi x)),$$

shown in Fig. 13, and use a time step, $\Delta t = 0.002$, which is small enough to eliminate time errors. The wave was propagated for one time period (i.e. two units) using a third-order Adams–Bashforth scheme. Four solution domains were considered for this test as shown in Fig. 14. Each domain spans the region $-1 \leq x \leq 1$ but is successively subdivided into equal lengths whilst maintaining the same aspect ratio. We can increase the expansion order on each domain to determine convergence with respect to expansion order and consider the error on each subdomain at constant expansion order to ascertain the convergence with respect to the number of elements. Fig. 15 shows the L_∞ error versus expansion order M for the four domains shown in Fig. 14. The L_∞ error is plotted on a logarithmic scale and we can therefore deduce spectral convergence from the linear

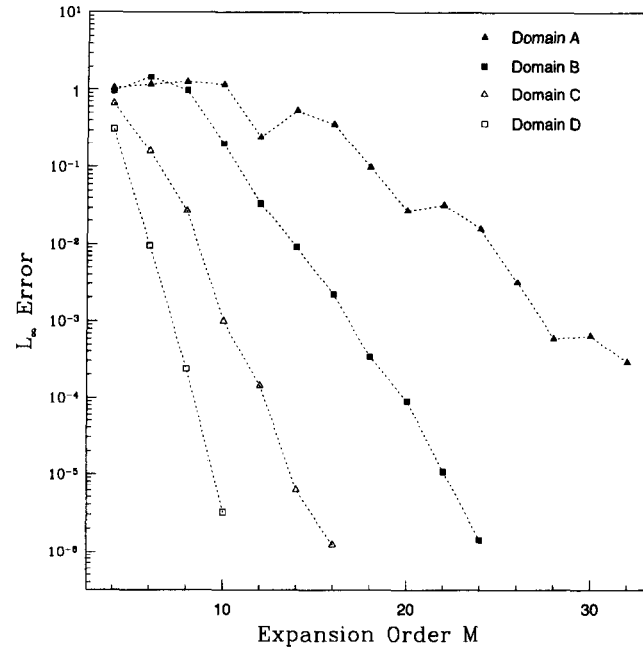


Fig. 15. L_∞ error with respect to expansion order M after one period. Each line corresponds to a different solution domain as shown in Fig. 14.

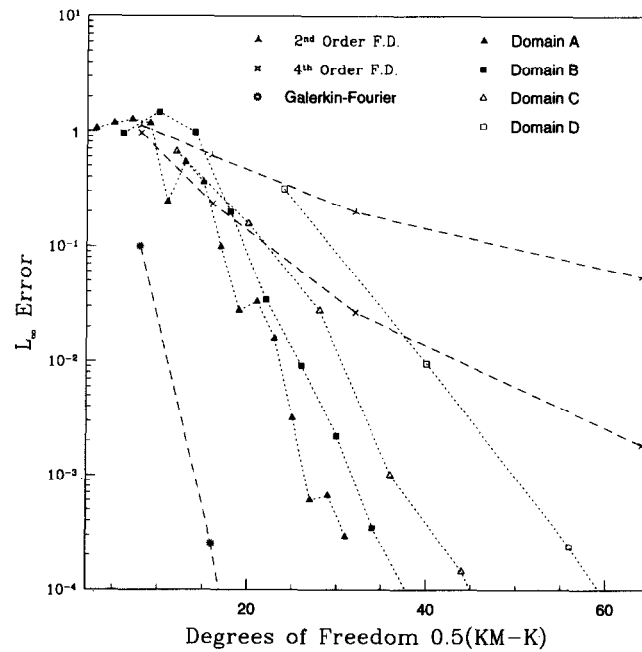


Fig. 16. L_∞ error versus degrees of freedom $\frac{1}{2}(KM - K)$ for all domains shown in Fig. 14. The points connected with dashed lines are results given in Canuto et al. [5] for Fourier-Galerkin and second- and third-order finite difference solution to this problem.

3.2. Helmholtz equation

We consider the elliptic Helmholtz equation in the form:

$$\nabla^2 u(x, y) - \mu u(x, y) = f(x, y) \quad (15)$$

where μ is a positive constant and the equation is supplemented with appropriate boundary condition which could be either Dirichlet, Neumann or a combination of both. We assume that $u(x, y) \in H_1$ and that $f(x, y) \in L_2$.

3.2.1. Discretization

Once again we consider the solution domain Ω divided into K triangular elements, T_k^2 , as shown in Fig. 7. We recall that the boundary of the solution domain is denoted by Γ and the boundary of each triangular element is denoted by ∂T_k^2 (see Section 3.1.1). Each triangular element is mapped to the standard triangular space where the local expansion base has already been explained as in Section 2. Within each element we can represent our local solution in the k^{th} element as:

$$u_M^k(x, y) = Pu^k(x, y) = \sum_{mn} \hat{u}^k g_{mn}^{1-2k}(r(x, y), s(x, y))$$

where r, s are the local co-ordinates and P is the projection operator. We represent the global approximation to the solution, U_M , as:

$$U_M = \sum_{k=1}^K u_M^k.$$

Substituting U_M into Eq. (15) we have:

$$\nabla^2 U_M - \mu U_M = f_M$$

where f_M is the global approximation to $f(x, y)$. Following the Galerkin philosophy we take the inner product of both sides of the equation with respect to the expansion basis $g_{mn}^{1-2k}(r, s)$ to obtain:

$$(g_{mn}^{1-2k}(r, s), \nabla^2 U_M) - \mu (g_{mn}^{1-2k}(r, s), U_M) = (g_{mn}^{1-2k}(r, s), f_M) \quad \forall (k; m, n).$$

Since $\nabla^2 U = \nabla \cdot \nabla U$ we can apply the divergence theorem to the first term so the equation becomes:

$$\int_{\Gamma} g_{mn}^{1-2k} \nabla U_M \cdot \mathbf{n} ds - (\nabla g_{mn}^{1-2k}, \nabla U_M) - \mu (g_{mn}^{1-2k}, U_M) = (g_{mn}^{1-2k}, f_M) \quad (16)$$

where $\mathbf{n}$ is the unit normal to the boundary Γ . We are only interested in the expansion modes which are zero on the solution boundaries, for the Dirichlet problem, so the inner product is only relevant with respect to those modes with undetermined coefficients. This is consistent with the notion that the Galerkin expansion satisfies the boundary conditions. The local solution can be represented at the quadrature points in matrix notation as:

$$\mathbf{u}_M = G\hat{\mathbf{u}}.$$

We wish to distinguish between the modes which are non-zero and zero on the solution boundary and so we will use G^i to represent the modes in the interior of the solution domain and G^b as the modes which have a non-zero value on the boundary of the solution domain. We should distinguish between these interior and boundary modes and the ones established in Section 2.1.2. The interior expansion modes for each local element will always be part of G^i . However, the boundary expansion modes for each element are not necessarily part of G^b . Only if the elemental boundary modes touch the solution boundary will they be part of G^b (i.e. when $\partial T_k^2 \cap \Gamma \neq \emptyset$) otherwise they will be part of G^i . This division of modes is equivalent to considering U_M to be split into two: a homogeneous component V_M and a boundary component W_M , where

$$V_M(\Gamma) = 0, \quad W_M(\Gamma) = U_M(\Gamma).$$

Since the inner product is with respect to modes that are zero on the boundary, Γ , the surface integral in equation (16) is zero. We can now represent an elemental version of Eq. (16) in matrix form as:

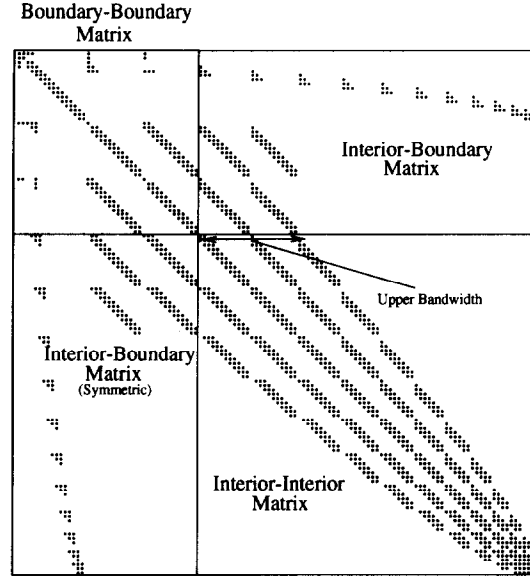


Fig. 17. The structure of the local Laplacian matrix on the standard triangle for $M = 10$. The matrix is symmetric and we note the banded structure of the interior–interior matrix.

$$-[(D_x G^i)^t W D_x G + (D_y G^i)^t W D_y G] \hat{u} - \mu (G^i)^t W G \hat{u} = (G^i)^t W f_M$$

where D_x and D_y denote a matrix form of the $\partial/\partial x$ and $\partial/\partial y$ operations. D_x and D_y can be constructed using the D_r and D_s matrices and the geometric factors which were mentioned in the previous section. Strictly speaking the f_M is a vector of approximations to the function $f(x, y)$ at the quadrature points in the local elements. We also know that $G = G^i + G^b$ where G^b corresponds to modes with known boundary coefficients. If we split each G matrix into the modes with unknown coefficients G^i and those with known coefficients G^b and put the known boundary points on the right hand side we get:

$$[(D_x G^i)^t W D_x G^i + (D_y G^i)^t W D_y G^i + \mu (G^i)^t W G^i] \hat{u}^i = - (G^i)^t W f_M - [(D_x G^i)^t W D_x G^b + (D_y G^i)^t W D_y G^b + \mu (G^i)^t W G^b] \hat{u}^b \quad (17)$$

where $\hat{u}^i$ and $\hat{u}^b$ are the unknown and known expansion coefficients, respectively. We have first to sum all local contributions into a global system and then solve the condensed global system. However, before we construct the global system we note that the third contribution to the left hand side matrix $(G^i)^t W G^i$ is the mass matrix discussed in Section 3.1.1. We also note that the structure of the first two parts to the left hand side matrix have a similar form to the mass matrix (see Fig. 17). This part of the matrix is the discrete Laplacian and when combined with the mass matrix, multiplied by μ , we have the stiffness matrix. The discrete Laplacian has an interior–interior matrix which is contained within the bandwidth of the mass matrix and thus the stiffness matrix has the same upper bandwidth as the mass matrix. This means that we can invert this system in an $O(M^3)$ type operation which is different from standard quadrilateral spectral elements where the cost is $O(M^4)$ for direct solvers [14].

If we let A_k, B_k be the left hand matrix and the right hand side vector of Eq. (17) in the k^{th} element, i.e.

$$A_k = (D_x G^i)^t W D_x G^i + (D_y G^i)^t W D_y G^i + \mu (G^i)^t W G^i$$

$$B_k = (G^i)^t W f_M - [(D_x G^i)^t W D_x G^b + (D_y G^i)^t W D_y G^b + \mu (G^i)^t W G^b] \hat{u}^b$$

then we can represent Eq. (17) as

$$A_k \hat{u}^i = B_k.$$

In a similar fashion, as shown in Section 3.1.1, we construct the global matrix and vector defined as:

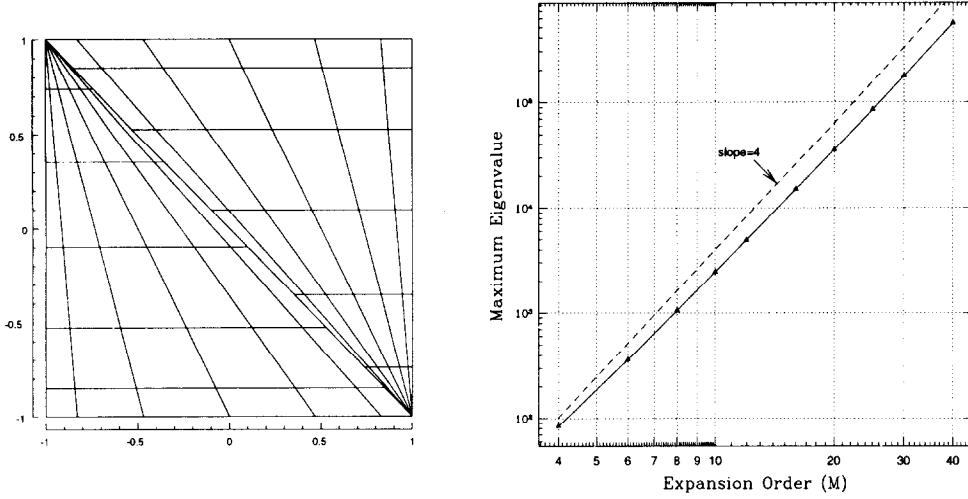


Fig. 18. The eigenspectrum was determined for the weak form of the diffusive operator for the periodic domain shown on the left. The growth rate of the absolute maximum eigenvalue with respect to expansion order M was determined as shown in the plot on the right.

$$A = \begin{bmatrix} A_1 & 0 & \cdots & 0 \\ 0 & A_2 & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & \cdots & A_K \end{bmatrix}, \quad B = \begin{bmatrix} B_1 \\ B_2 \\ \cdots \\ B_K \end{bmatrix}.$$

Now using the local to global mapping matrix Z we can construct the global matrix system which is:

$$Z^T A Z \hat{u}_g = Z^T B$$

where $\hat{u}_g$ is a vector of the global expansion coefficients. We can therefore determine the solution to the expansion coefficients by inverting $Z^T A Z$ so

$$\hat{u}_g = (Z^T A Z)^{-1} Z^T B$$

and the expansion in physical space can be obtained by performing a backwards transformation.

3.2.2. Spectrum of the weak diffusive operator

We next investigate growth of the largest eigenvalue of the weak diffusive operator. Consider the two-dimensional parabolic equation:

$$\frac{\partial u(x, y; t)}{\partial t} = \nabla^2 u(x, y; t). \quad (18)$$

If we spatially discretize this equation we get a semi-discrete system of the form:

$$\frac{d\hat{u}_g(t)}{dt} = (Z^T M Z)^{-1} Z^T A Z \hat{u}_g(t)$$

where M is the summation of the local mass matrices as defined in (8) and A is the matrix as defined in the last section with $\mu = 0$. For stability considerations we require that the spectrum of $(Z^T M Z)^{-1} Z^T A Z$ is contained within the stability region of the time stepping scheme used. If we consider a periodic domain as shown in Fig. 18 and determine the spectrum of $(Z^T M Z)^{-1} Z^T A Z$ for this domain we find that the maximum absolute eigenvalue varies with M in a manner shown by the plot on the right in Fig. 18.

The growth rate in this case was ≈ 3.8 . This can be appreciated from the plot in Fig. 18 which compares the points with a line of slope 4 as indicated by the dashed line.

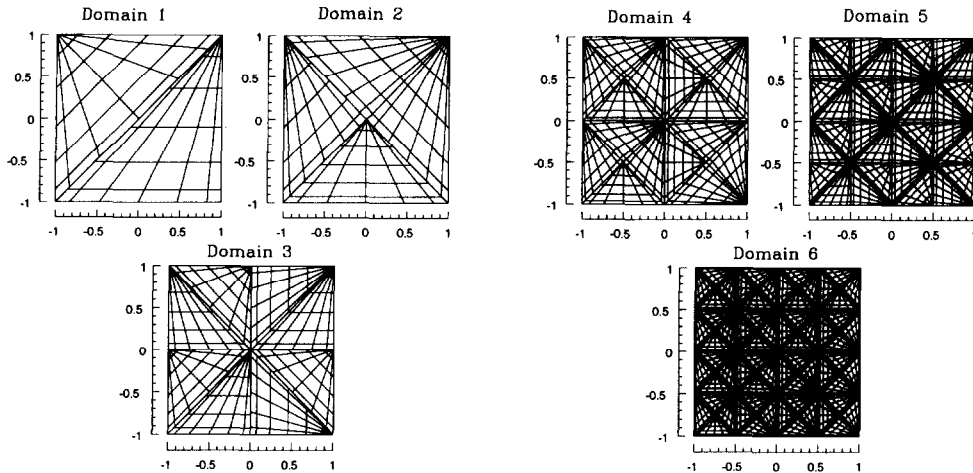


Fig. 19. Solution domains for Helmholtz test problem. Each domain spans the region $-1 \leq x \leq 1$, $-1 \leq y \leq 1$ but is divided into 2, 4, 8, 16, 32 or 64 equal triangular elements. The intersecting lines denote quadrature points.

3.2.3. Results and validation

The implementation and computational complexity of solving these problems is addressed in [24]. The main operations are inverting the stiffness matrix using static condensation, conditioning of the global matrix systems and evaluation of the backward transformation and inner product using sum factorization. The most significant result is that all these tasks can be performed in $O(M^3)$ operations per element. This is asymptotically better than standard quadrilateral spectral elements. For $M \approx 10$ both methods have comparable computation expense.

The initial test we shall consider is a Helmholtz problem with a coefficient of $\mu = 1$ having a solution of the form:

$$u(x, y) = \sin(\pi x) \cos(\pi y).$$

Once again we shall consider the solution to this problem on a variety of domains using Dirichlet boundary condition at different expansion orders. The absolute solution domain spans the area given by $-1 \leq x \leq 1$, $-1 \leq y \leq 1$ but is subdivided into 2, 4, 8, 16, 32 or 64 equal triangular elements. We will refer to these as domains 1–6 as shown in Fig. 19. For each domain the Helmholtz problem was solved for a variety of expansion orders M and the L_∞ and H_1 errors versus expansion order are shown in Fig. 20. Spectral convergence is demonstrated by the fact that on this lin-log plot the convergence lines are slightly better than linear. The error is resolved to machine precision which is of the order of 10^{-13} – 10^{-14} for these norms. Finally, we note that if we consider a line of constant expansion order we see convergence with the number of elements. The lines corresponding to domains 2 and 3 as well as 4 and 5 are closely spaced and this is not too surprising when we note that the resolution length along the $\pm 45^\circ$ lines in domains 2 and 3 as well as 4 and 5 are the same.

The second test we wish to consider is one which involves a variety of triangular elements of different aspect ratios and orientations. Such a domain is shown in Fig. 21 and Fig. 22. The Dirichlet–Helmholtz problem with $\mu = 1$ and $u(x, y) = \sin(\pi x) \cos(\pi y)$ was solved on this domain and the L_∞, L_2 errors were plotted with respect to expansion order as can also be seen in Fig. 21. Spectral convergence is maintained in this more complicated domain even though we have used a random discretization which is not a Delaunay triangularization.

4. Solution of the incompressible Navier–Stokes equations

The following formulation has been implemented in the new spectral element flow code *NekTar* which is part of a new generation of incompressible Navier–Stokes solvers.

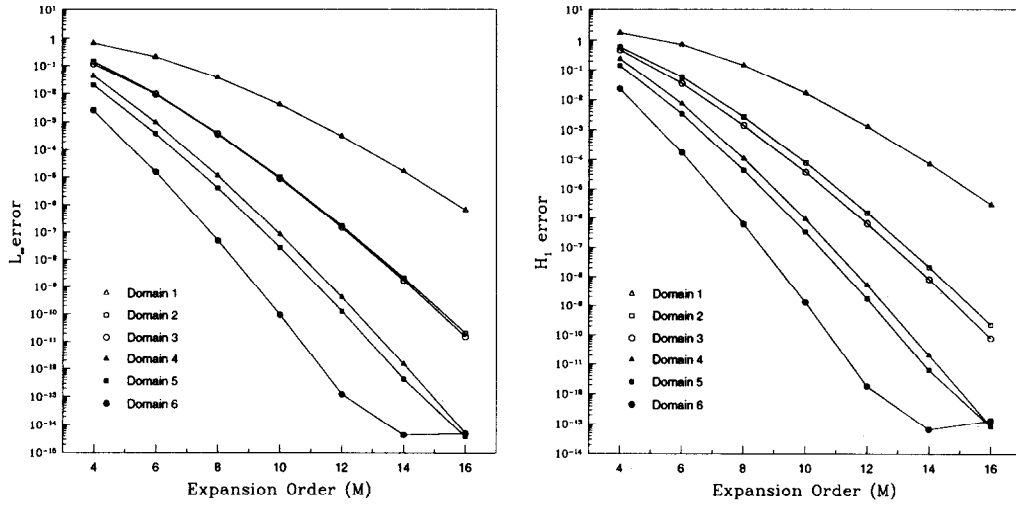


Fig. 20. Convergence plots in the L_∞ and H_1 norms for a Helmholtz problem with $\mu = 1$ versus expansion order M . The solution is obtained on different domains, shown in Fig. 19, which spanned the same area in x, y .



Fig. 21. Solution to the Helmholtz problem with solution $u(x, y) = \sin(\pi x) \cos(\pi y)$ and $\mu = 1$.

4.1. Formulation

We wish to apply the previous formulations for the solution of the advection and Helmholtz problems to solve the Navier–Stokes equations. We are going to consider the two-dimensional incompressible Navier–Stokes equations which can be written:

$$\frac{\partial \mathbf{V}}{\partial t}(x, y; t) + (\mathbf{V}(x, y; t) \cdot \nabla) \mathbf{V}(x, y; t) = -\nabla p(x, y; t) + \nu \nabla^2 \mathbf{V}(x, y; t),$$

$$\nabla \cdot \mathbf{V}(x, y; t) = 0.$$

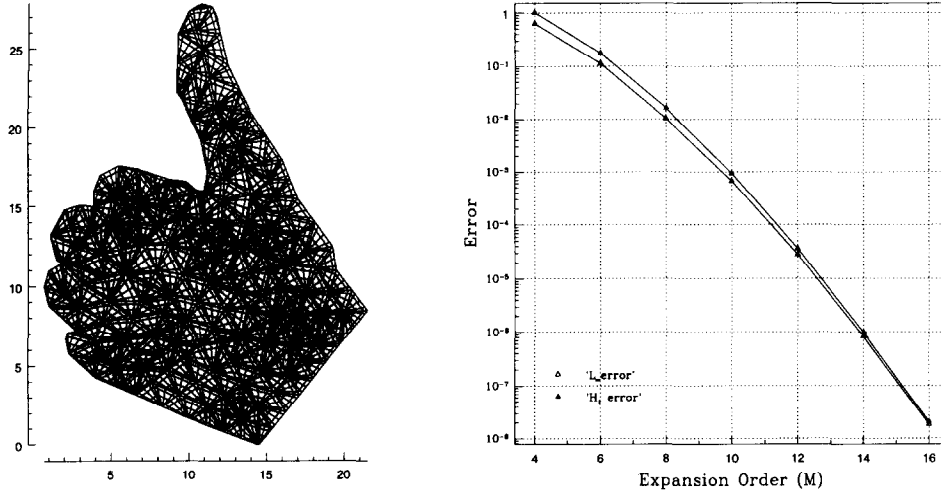


Fig. 22. Computational mesh where the intersecting lines denote the quadrature points (left). Convergence to the Helmholtz problem with solution $u(x, y) = \sin(\pi x) \cos(\pi y)$ (right). Spectral convergence is obtained independently of the complexity of the discretization.

V denotes the velocity of the fluid with components $V = [u(x, y; t), v(x, y; t)]$ in the x and y direction; $p(x, y; t)$ is the pressure and ν is the kinematic viscosity. To discretize these equations we have chosen to use the high-order splitting scheme [15]. This scheme propagates the fields $V^n = V(x, y; t^n)$, $p^n = p(x, y; t^n)$ over a time step Δt to determine the fields V^{n+1} , p^{n+1} in the three substeps:

$$\frac{\hat{V} - V^n}{\Delta t} = \sum_{q=0}^{J_e-1} \beta_q N(V^{n-q}), \quad (19a)$$

$$\frac{\hat{V} - \hat{V}}{\Delta t} = -\nabla \bar{p}^{n+1}, \quad (19b)$$

$$\frac{V^{n+1} - \hat{V}}{\Delta t} = \nu \sum_{q=0}^{J_l-1} \gamma_q L(V^{n+1-q}), \quad (19c)$$

where

$$N(V) = (V \cdot \nabla) V, \quad L(V) = \nabla^2 V,$$

$$\nabla \bar{p}^{n+1} = \frac{1}{\Delta t} \int_{t^n}^{t^{n+1}} \nabla p \, dt.$$

Here $N(V)$ and $L(V)$ are the non-linear and linear terms, respectively, and $\bar{p}$ can be considered as a scalar field which ensures that the final velocity field is incompressible at time level $(n+1)$; β and γ are time integration constants. We shall consider each substep (19a), (19b) and (19c) separately and give a brief explanation as how they are evaluated.

Step (19a) is an advection operator. However, this equation now contains the non-linear product $(V \cdot \nabla) V$. This non-linearity makes it difficult to discretize this step in an implicit fashion in time. Nevertheless, as we saw in Section 3.1 the nature of our scheme is such that the explicit treatment of convective operators is not prohibitive. We might interpret the non-linear convective operator as a linear convective operator with an instantaneous wave speed of V^{n-q} . It is therefore apparent that this step is very similar to the advection equation considered in Section 3.1 where we also considered the wave speed V as being divergent free. The major distinction here is that we have to evaluate the non-linear product. This product is easily evaluated in physical space; we recall that the projection operator involves the use of an evaluation operator at the quadrature points in a collocation type operation. Thus the non-linear product is evaluated at the quadrature points as the direct product of the local velocity values with the relevant partial derivatives. The inner product of this term can then be found to form the right hand side of Eq. (19a).

We have used the Adams–Bashforth multistep time stepping scheme to evaluate these terms and so for the third-order time accurate scheme ($J_e = 3$) the time integration constants are $\beta_0 = 23/12$, $\beta_1 = -16/12$ and $\beta_2 = 5/12$. Strictly speaking to evaluate the right hand side inner product $(\hat{g}_{mn}^{1-2}, (V \cdot \nabla)V)$ exactly requires the integration order to be raised because $(V \cdot \nabla)V$ belongs in G_{2M-1} if V is expanded in G_M . Nevertheless, no adverse effects have been noted when this term has been treated as an expansion in G_M [22,25,28]. During this step no boundary conditions are imposed and so all unknown boundary points are treated as a type of outflow boundaries.

If we take the divergence of step (19b) and impose the condition that $\hat{V}$ be divergence free (i.e. $\nabla \cdot \hat{V} = 0$) we obtain the Poisson equation:

$$\nabla^2 \bar{p} = \nabla \cdot \left(\frac{\hat{V}}{\Delta t} \right). \quad (20)$$

This equation needs to be augmented with appropriate boundary conditions, the choice of which is important in minimizing the splitting error [15]:

$$\frac{\partial \bar{p}^{n+1}}{\partial n} = \mathbf{n} \cdot \left[- \sum_{q=0}^{J_e-1} \beta_q N(V^{n-q}) - \nu \sum_{q=0}^{J_e-1} \beta_q [\nabla \times (\nabla \times V^{n-q})] \right]$$

where $\mathbf{n}$ is the unit normal to the boundary Γ . Having evaluated the boundary conditions using the information from the previous time steps we can solve Eq. (20) to determine $\bar{p}^{n+1}$ as a Helmholtz equation with $\mu = 0$ as explained in Section 3.2.

We wish to use the Crank–Nicolson time stepping scheme in the final step (19c) which requires $J_i = 2$, $\gamma_0 = \gamma_1 = 1/2$. This step (19c) can be formulated in terms of a θ scheme which reduces to the Crank–Nicolson scheme for $\theta = 0.5$ and the Euler backwards scheme for $\theta = 0$. So the last step can be written as:

$$(1 - \theta) \nabla^2 V^{n+1} - \frac{1}{\nu \Delta t (1 - \theta)} V^{n+1} = - \frac{2}{\nu \Delta t} \hat{V} - \theta \nabla^2 V^n \quad (21)$$

where

$$\hat{V} = \hat{V} - \Delta t \nabla \bar{p}^{n+1}.$$

This is a Helmholtz equation with $\mu = 2/(\nu \Delta t (1 - \theta))$. Boundary conditions for this equation are specified by along the boundary, Γ , as either Dirichlet or Neumann conditions. In order to eliminate both an implicit and explicit evaluation of the Laplacian we can introduce the averaged variable V^* defined as:

$$V^* = (1 - \theta) V^{n+1} + \theta V^n$$

and substituting this into Eq. (21) we obtain:

$$\nabla^2 V^* - \frac{1}{\nu \Delta t (1 - \theta)} V^* = - \frac{\hat{V} + \frac{\theta}{1 - \theta} V^n}{\nu \Delta t}.$$

Having determined V^* we can recover V^{n+1} using V^n since

$$V^{n+1} = \frac{V^* - \theta V^n}{1 - \theta}.$$

4.2. Results

4.2.1. Moffatt flow

The first result we simulate is viscous flow near a sharp corner at zero Reynolds number (Stokes flow). We consider a wedge which has an aspect ratio of 2 : 1 as shown in Fig. 23 which means that the apex angle is $\approx 28.1^\circ$. The wedge was discretized into 30 triangular elements and expansion bases of $M = 9, 12, 15$ and 18

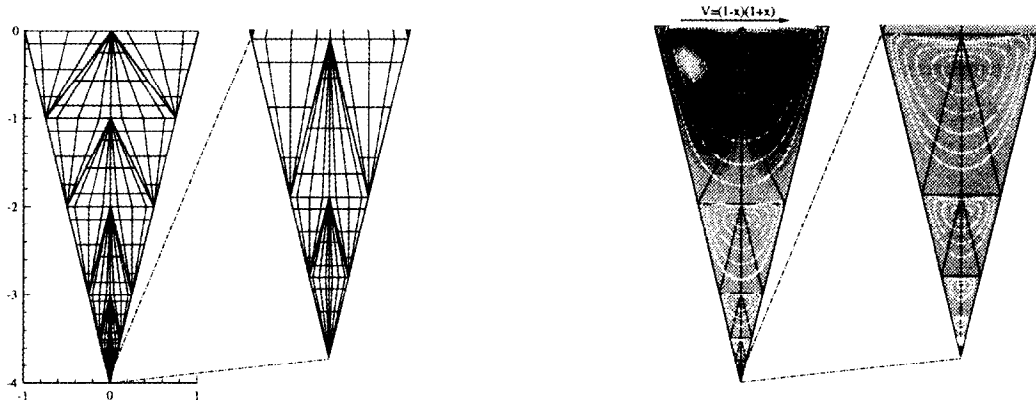


Fig. 23. A wedge with an aspect ratio of 2 : 1 was discretized using 30 elements and an expansion order of $M = 18$ as shown on the left. Stokes flow was then solved in this domain driven by a parabolic forcing velocity. At steady state nine eddies were observed as indicated by the streamline plot on the right (there are three eddies in the last two elements).

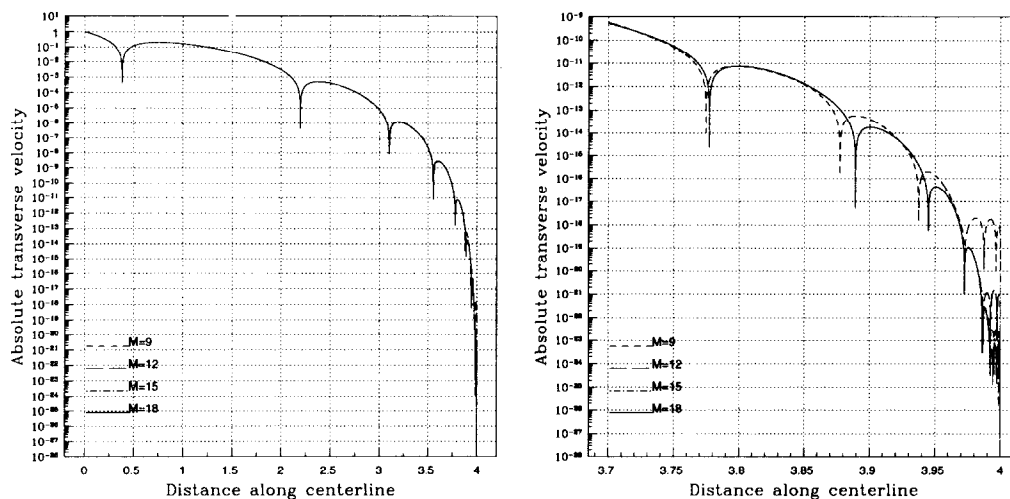


Fig. 24. The centerline transverse velocity as a function of perpendicular height from the top of the wedge shown in Fig. 23.

were used to determine the steady state solution using the Euler backwards scheme (i.e. $\theta = 0$) with a time step of $\Delta t = 1 \cdot 10^{-4}$. The flow was driven by a parabolic forcing velocity along the top of the wedge which has a maximum value of one. Having allowed the solution to reach a steady state we find that at the highest resolution considered ($M = 18$) we can resolve nine eddy formulations as indicated by the streamline plot in Fig. 23.

Using a similarity solution Moffatt [29] derived an asymptotic result for the strength and location of an “infinite” number of eddies in Stokes flow near sharp corners. The relations were dependent on the wedge angle; for our wedge angle (28.1°) it is predicted that the strength of each eddy should asymptotically (i.e. away from the forced top section) be about 406 times weaker than the previous eddy. Moffatt’s measure of the ‘intensity’ of consecutive eddies was the ratio of the local maximum transverse velocity along the centerline. Therefore, if we take a profile along the centerline of our solution and plot the transverse (x -direction) velocity as a function of perpendicular distance from the top of wedge we obtain the distributions shown in Fig. 24 for the four resolutions $M = 9, 12, 15$ and 18 . (Here we have used log of the absolute velocity on the y axis since the eddies decay so rapidly.)

At the center of an eddy we expect the transverse velocity to be zero which is indicated by the spikes in Fig. 24. After each spike we note that there are local maxima which are tabulated in Table 1. Using these values we can determine the ratio of maximum velocities in order to evaluate Moffatt’s eddy ‘intensity’. These values are also recorded in Table 1. We note from Fig. 24 that at all resolutions the first four eddies have been resolved to the accuracy of the plot. The $M = 9$ run has resolved the fifth-seventh eddy although they appears

Table 1

Values of maximum transverse velocity along the centerline of Stokes solution in a wedge shown in Fig. 23 using expansion orders $M = 9, 12, 15$ and 18 . The highest value corresponds to the largest eddy near to the top of the wedge. Also shown is the ratio of relative velocity as a measure of eddy 'intensity'. For this case Moffatt predicted an asymptotic ratio of 406

Eddy	Expansion order							
	$M = 9$		$M = 12$		$M = 15$		$M = 18$	
	Max. velocity	Ratio	Max. velocity	Ratio	Max. velocity	Ratio	Max. velocity	Ratio
1	$-2.012175 \cdot 10^{-1}$		$-2.009948 \cdot 10^{-1}$		$-2.010064 \cdot 10^{-1}$		$-2.010063 \cdot 10^{-1}$	
		401.26		400.28		400.31		400.31
2	$5.014660 \cdot 10^{-4}$		$5.021325 \cdot 10^{-4}$		$5.021297 \cdot 10^{-4}$		$5.021298 \cdot 10^{-4}$	
		398.35		406.63		406.72		406.72
3	$-1.258846 \cdot 10^{-6}$		$-1.234856 \cdot 10^{-6}$		$-1.234572 \cdot 10^{-6}$		$-1.234571 \cdot 10^{-6}$	
		458.10		406.71		406.72		406.72
4	$2.747955 \cdot 10^{-9}$		$3.036194 \cdot 10^{-9}$		$3.035402 \cdot 10^{-9}$		$3.035399 \cdot 10^{-9}$	
		370.72		406.72		406.72		406.72
5	$-7.412465 \cdot 10^{-12}$		$-7.465022 \cdot 10^{-12}$		$-7.463042 \cdot 10^{-12}$		$-7.463033 \cdot 10^{-12}$	
		127.26		406.74		406.72		406.72
6	$5.824836 \cdot 10^{-14}$		$1.835340 \cdot 10^{-14}$		$1.834913 \cdot 10^{-14}$		$1.834910 \cdot 10^{-14}$	
		286.76		406.74		406.72		406.72
7	$-2.031242 \cdot 10^{-16}$		$-4.512268 \cdot 10^{-17}$		$-4.511441 \cdot 10^{-17}$		$-4.511432 \cdot 10^{-17}$	
				414.33		406.75		406.72
8	unresolved		$1.089041 \cdot 10^{-19}$		$1.109149 \cdot 10^{-19}$		$1.109229 \cdot 10^{-19}$	
						398.75		402.25
9	unresolved		unresolved		$-2.781580 \cdot 10^{-22}$		$-2.757530 \cdot 10^{-22}$	

to still be converging to their final values. Both the $M = 12$ and $M = 15$ runs resolve the fifth-eighth eddies and the appear to be converged to the accuracy of the plot. The ninth eddy is not resolved by the $M = 12$ run but is quite well resolved by the $M = 15$ run. The $M = 18$ run resolves the ninth eddy pretty well although there is no definite appearance of the tenth eddy. This is perhaps understandably since there are now three eddies in the last two elements and to capture further eddies would require h type refinement in these two elements.

A more informative measure of convergence is shown in Table 1. As mentioned the first four eddies were resolved by all expansion orders and this is substantiated by the fact that the measured values for the first three eddies are within 1.5% of each other. The fourth eddy was not completely captured by the $M = 9$ case and this is shown by the ratio between the third-fourth eddy at this resolution. All the other runs have captured the eddies very accurately and there is a definite convergence to ratio value of the 406.72. The eighth-ninth eddies are only captured by the $M = 15$ and $M = 18$ runs and the ratio has not quite converged at $M = 18$ although it is within 1.0%.

4.2.2. Wannier flow

Wannier flow is Stokes flow past a rotating circular cylinder next to a moving wall. The exact solution due to Wannier [30] allows us to evaluate the error in a domain involving curvi-linear elements. The exact solution can be written:

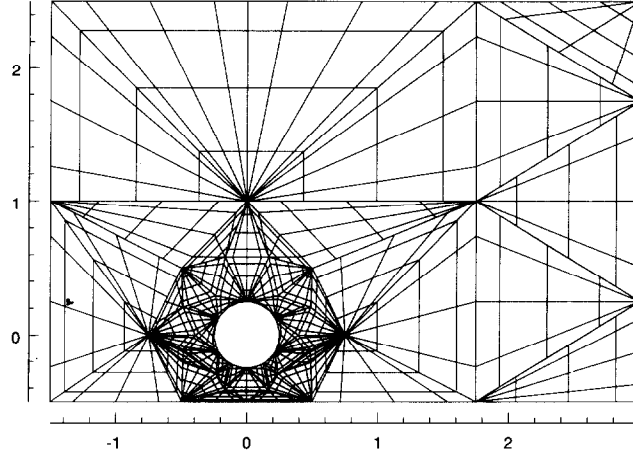


Fig. 25. Discretized solution domain for the Wannier-Stokes flow using 65 elements. The intersecting lines within each element indicate the quadrature points for $M = 5$.

$$\begin{aligned}
 u(x, y_1 = y + d) &= -\frac{2(A + Fy_1)}{K_1} \left[(s + y_1) + \frac{K_1}{K_2}(s - y_1) \right] - F \ln \left(\frac{K_1}{K_2} \right) \\
 &\quad - \frac{B}{K_1} \left[s + 2y_1 - \frac{2y_1(s + y_1)(s + y_1)}{K_1} \right] \\
 &\quad - \frac{C}{K_2} \left[s - 2y_1 + \frac{2y_1(s - y_1)(s - y_1)}{K_2} \right] - D \\
 v(x, y_1 = y + d) &= \frac{2x}{K_1 K_2} (A + Fy_1)(K_2 - K_1) - \frac{2Bxy_1(s + y_1)}{K_1^2} - \frac{2Cxy_1(s - y_1)}{K_2^2}
 \end{aligned}$$

where we define:

$$\begin{aligned}
 A &= -\frac{Ud}{\ln(\Gamma)} + \frac{d\omega}{2s}, \quad B = \frac{2U(d + s)}{\ln(\Gamma)} + \frac{(d + s)\omega}{s}, \quad C = \frac{2U(d - s)}{\ln(\Gamma)} + \frac{(d - s)\omega}{s}, \quad D = -U, \quad F = \frac{U}{\ln(\Gamma)}, \\
 K_1 &= x^2 + (s + y_1)^2, \quad K_2 = x^2 + (s - y_1)^2, \quad s^2 = d^2 + R^2, \quad \Gamma = \frac{d + s}{d - s}.
 \end{aligned}$$

Here we have used a cylinder of radius $R = 0.25$ which is a distance $d = 0.5$ from the moving wall. The wall is moving with a velocity of $U = 1$ and the cylinder is rotating in a counter clockwise sense with an angular velocity of $\omega = 2$. The domain was split into 65 elements and the discretized domain is shown in Fig. 25.

As can be seen on this mesh we have curvi-linear elements surrounding the cylinder. The intersecting lines within each element indicate the quadrature points for $M = 5$. Here a varied triangularization is introduced in order to test the convergence of the method on distorted meshes. These lines are blended from one edge to the others and so in the curvi-linear elements they are also deformed. All elements are mapped to the standard triangle when integration is performed and because of their deformed nature the Jacobian is not constant within curvi-linear elements. As mentioned previously this non-constant Jacobian destroys the sparsity of the interior-interior coupling matrices in the mass and stiffness matrices. Nevertheless, since there are only a few of these elements performance is not noticeably affected. The steady state solution with Dirichlet boundary conditions using an expansion basis of $M = 11$ is shown in Fig. 26. This figure shows isocontours of absolute total velocity as well as streamlines of the steady flow. We note the recirculation region captured at the top of the rotating cylinder.

As a final validation we computed the L_∞ and H_1 error of the solution at steady state for various expansion orders as shown in Fig. 27. We note that the spectral convergence of this flow as indicated by the near linear slope of the lines.

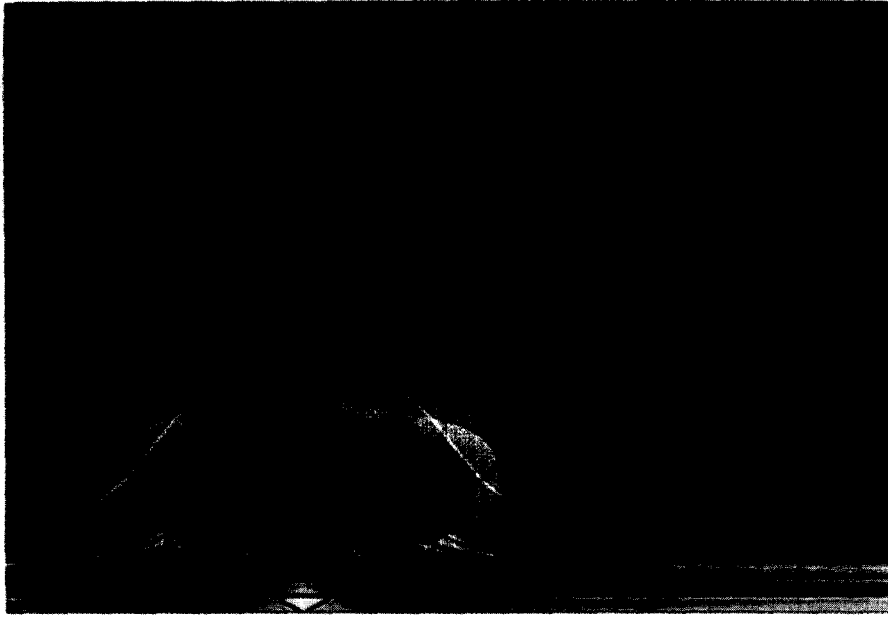


Fig. 26. Steady state solution of the Wannier–Stokes flow using an expansion order of $M = 11$. The isocontours show x -component velocity and the white lines represent streamlines.

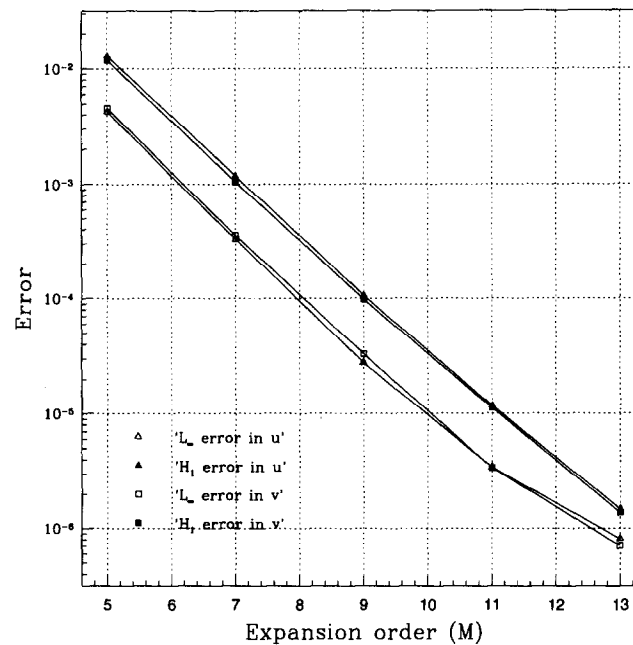


Fig. 27. Plots of the L_∞ and H_1 errors as a function of expansion order for the u and V velocity field of the steady state solution to the Wannier–Stokes flow.

4.2.3. Kovaszny flow

The first Navier–Stokes solution we shall consider is the Kovaszny flow. This is a laminar flow behind a two-dimensional grid, the exact solution of which is due to Kovaszny [31]. This solution can be written as a function of Reynolds number Re in the form:

$$u(x, y) = 1 - e^{\lambda x} \cos(2\pi y)$$

$$v(x, y) = \frac{\lambda}{2\pi} e^{\lambda x} \sin(2\pi y)$$

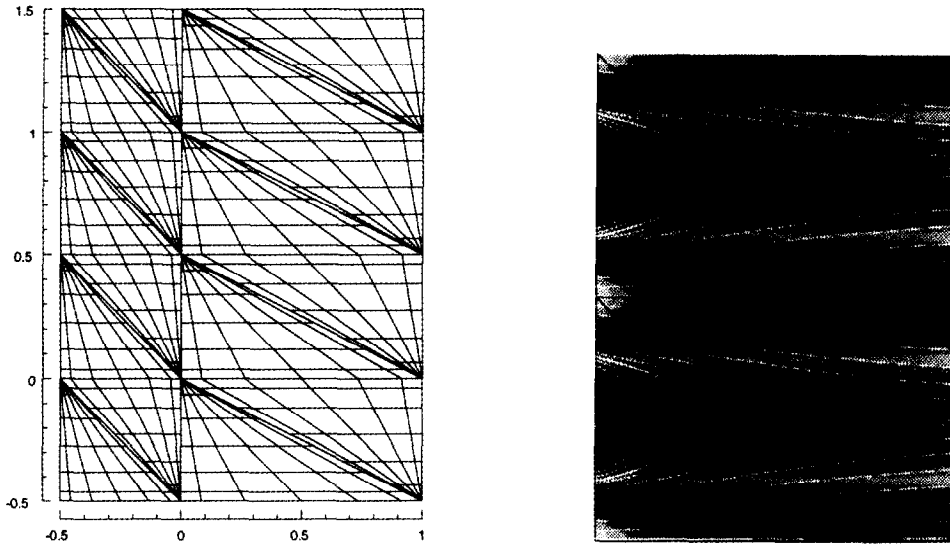


Fig. 28. Domain discretization and steady state solution for the Kovasznay flow at Reynolds number $Re = 40$. The solutions shown are the steady state stream lines.

where

$$\lambda = \frac{Re}{2} - \left(\frac{Re^2}{4} + 4\pi^2 \right)^{\frac{1}{2}}.$$

Using the exact solution as Dirichlet boundary conditions, a steady state solution was obtained using the discretization shown in Fig. 28. Also shown in this figure are the steady state streamlines and isocontours of the x -component velocity at a Reynolds number of $Re = 40$ using an expansion basis of $M = 7$.

Once again knowing the exact solution allows us to calculate the convergence with expansion order and this is shown in Fig. 29. In this figure we see the L_∞ and H_1 errors as a function of expansion basis. The convergence is spectral and we note that double precision accuracy is achieved for this smooth solution.

4.2.4. Flow past a circular cylinder

For the final example we consider a flow which demonstrates the different factors illustrated in the previous simulations. To this end we consider flow past a circular cylinder at Reynolds number of $Re = 100, 200$ and 300 . This case clearly involves a complex geometry and is unsteady and two-dimensional at this Reynolds number. As shown in Fig. 30 we consider a very large computational domain so that we can impose free-stream boundary conditions at the top and bottom of the domain without noticeably affecting the solution. We impose a free stream inflow velocity field of $\mathbf{V} = [1, 0]$ and use a zero-Neumann outflow boundary condition downstream of the cylinder. The mesh in Fig. 30 demonstrates the flexibility of the triangular elements in discretizing the solution domain. In the far field where the solution is relatively smooth we can use large element whereas within the wake region immediately behind the cylinder we can cluster more elements to achieve greater accuracy.

Fig. 31 shows isocontours of vorticity for this test. At these Reynolds numbers we see time-periodic vortex shedding in the wake behind the cylinder. Using Hammache and Gharib's [32] accurate experimental fit for the Strouhal frequency as function of Reynolds number we find that at $Re = 100, 200$ and 300 the Strouhal frequency should be $St_o = 0.1658, 0.1990$ and 0.2200 , respectively. We observe in the simulations Strouhal frequencies of $St_o = 0.1667, 0.1978$ and 0.2167 for these Reynolds numbers which are with 0.5% , 0.6% and 5% , respectively. The last frequency at $Re = 300$ is less accurate due to the introduction of three dimensional effects at this Reynolds number.

The cost of this computation is comparable to the computations performed with quadrilateral spectral element methods [33]. Nevertheless, direct comparison of these two methods is complicated by the different expansion bases used and the relative geometric flexibility of each method. This issue is being addressed in current work.

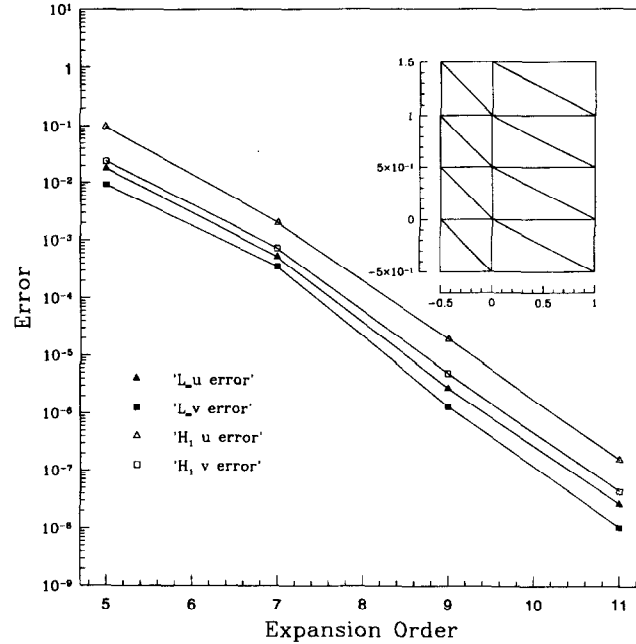


Fig. 29. Convergence in the L_∞ and H_1 norms as a function of expansion order for the steady state Kovaszny flow at Reynolds number $Re = 40$.

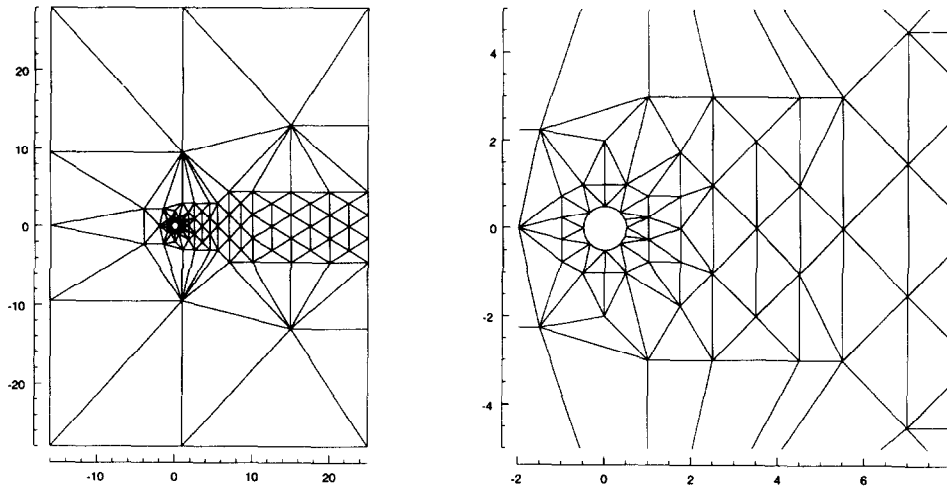


Fig. 30. Solution domain and discretization for the circular cylinder calculation. The domain is split into 173 elements as shown on the left. The triangular elements allow the mesh to be easily clustered behind the cylinder where greater resolution is required as shown by the mesh on the right.

5. Conclusions

In this paper we have developed a new hierarchical modal basis for triangular spectral (hp) elements. A complete analysis of the advection and Helmholtz equations as well as the formulation of a Navier–Stokes solver, *NekTar*, were presented. The three-dimensional formulation is currently under development and will also be part of *NekTar*. This will represent the next generation of spectral element solvers on unstructured meshes.

The new discretization uses standard triangular finite volume meshes since all that is required is a conforming triangular discretization. Our experiments suggest that exponential convergence is obtained even for very distorted meshes as demonstrated by the ‘thumb’ mesh and the Wannier flow mesh where arbitrary shaped triangles were intentionally employed. While Delaunay triangulization may provide an automatic procedure in

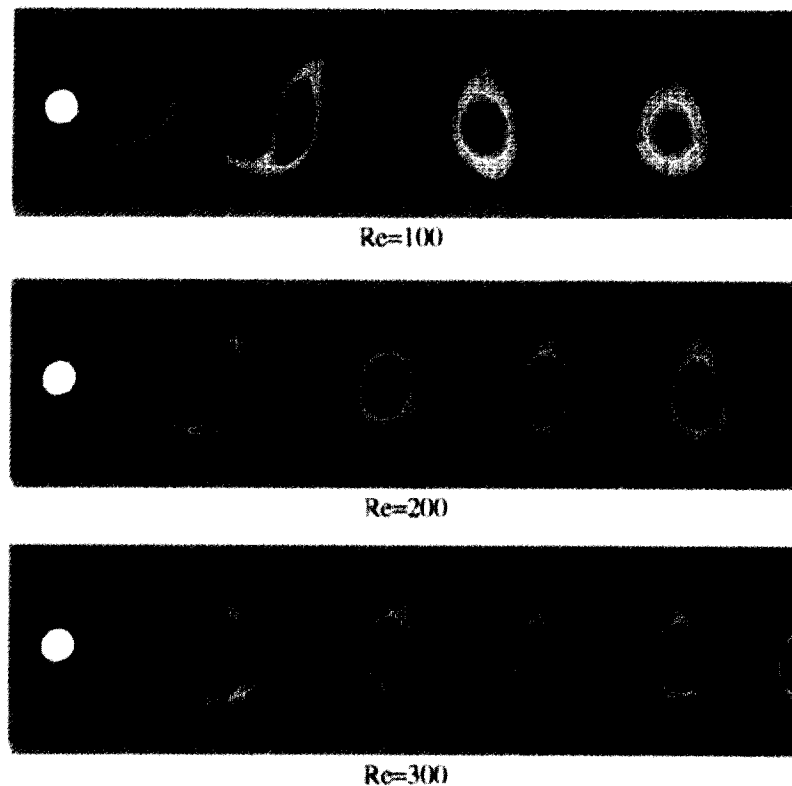


Fig. 31. Instantaneous isocontours of vorticity for flow over a circular cylinder at Reynolds numbers of $Re = 100, 200$ and 300 . The calculation was performed with an expansion order of $M = 11$.

generating unstructured meshes, unlike other unstructured methods, such a quality is not necessary. Adaptive discretization and composite gridding for *surgical refinement* in flows with localized structure, e.g. vortices and near-wall turbulent streaks, is a built-in feature of the method as variable modal expansions can be employed on each element, solely within the interior or just along an elemental edge.

With regard to computational efficiency, the generalized tensor product of the new basis which allows use of sum factorization techniques makes the new code comparable to standard quadrilateral spectral and p-type finite elements. In fact, the operation count for global matrix inverses is of $O(N^3)$ which differs from standard quadrilateral spectral elements where the cost is $O(N^4)$ for [14]. Therefore, as the number of modes increases the new method is asymptotically more efficient.

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References

- [1] T.J. Baker, Developments and trends in three-dimensional mesh generation, *Appl. Numer. Math.* 5 (1989) 275.
- [2] N.P. Weatherill and O. Hassan, Efficient three-dimensional Delaunay triangularization with automatic point creation and imposed boundary constraints, *Int. J. Numer. Methods Engrg.* (1994) 37–2005.
- [3] G.E. Karniadakis and S.A. Orszag, Nodes, modes and flow codes, *Phys. Today* 46(3) (1993) 34.
- [4] D. Gottlieb and S.A. Orszag, *Numerical analysis of spectral methods: Theory and applications* (SIAM-CMBS, Philadelphia, PA, 1977).
- [5] C. Canuto, M.Y. Husanini, A. Quateroni and T.A. Zang, *Spectral Methods in Fluid Dynamics* (Springer-Verlag, Berlin, 1988).

- [6] T.J. Barth, Recent Developments in high order k -exact reconstruction on unstructured meshes, AIAA-93-0668.
- [7] L. Demkowicz, J.T. Oden, W. Rachowicz and O. Hardy, Towards a universal h-p adaptive finite element strategy, part 1: constrained approximation and data structure, *Comput. Methods Appl. Mech. Engrg.* 77 (1989) 79.
- [8] R.A. Shapiro and E.M. Murman, Higher-order and 3-d finite element methods for the Euler equations, AIAA-89-0655.
- [9] M. Dubiner, Spectral methods on triangles and other domains, *J. Sci. Comput.* 6 (1991) 345.
- [10] A.T. Patera, A spectral method for fluid dynamics: Laminar flow in a channel expansion, *J. Comput. Phys.* 54 (1984) 468.
- [11] G.E. Karniadakis E.T. Bullister and A.T. Patera, A spectral element method for solution of two- and three-dimensional time dependent Navier–Stokes equations, in: P.G. Bergen, K.J. Bathe and W. Wunderlich, eds., *Finite Element Methods for Nonlinear Problems* (Springer-Verlag, Berlin, 1985) 803.
- [12] G.E. Karniadakis, Spectral element simulation of laminar and turbulent flows in complex geometries, *Appl. Numer. Math.* 6 (1989) 85.
- [13] G. Anagnostou, Nonconforming Sliding Spectral Element Methods for the Unsteady Incompressible Navier–Stokes Equations. PhD Thesis, Massachusetts Institute of Technology, 1990.
- [14] R.D. Henderson and G.E. Karniadakis, Unstructured spectral element methods for simulation of turbulent flows, *J. Comput. Phys.*, submitted for publication.
- [15] G.E. Karniadakis, M. Israeli and S.A. Orszag, High-order splitting methods for the incompressible Navier–Stokes equations, *J. Comput. Phys.* 97 (1991) 414.
- [16] J.T. Oden, W. Wu and V. Legat, An hp adaptive strategy for finite element approximation of the Navier–Stokes equations, in: *Finite Elements in Fluids—New trends and applications* (Pineridge Press, Swansea, 1993) 32.
- [17] A. Peano, Hierarchies of conforming finite elements for plane elasticity and plate bending, *Comput. Maths. Appl.* 2 (1976) 221.
- [18] I. Babuska and M. Suri, The p and h-p versions of the finite element method: Basic principles and properties, Technical Report, Institute for Physical Science and Technology, University of Maryland, 1992.
- [19] I. Babuska, B.A. Szabo and I.N. Katz, The p-version of the finite element method. *SIAM J. Numer. Anal.* 18 (1981) 515.
- [20] J.T. Oden, Optimal hp-finite element methods. Technical Report TICOM Report 92-09, University of Texas at Austin, 1992.
- [21] M. Abramowitz and I.A. Stegun, *Handbook of Mathematical Functions* (Dover, New York, 1970).
- [22] Y. Maday and E.M. Ronquist, Optimal error analysis of spectral methods with emphasis on non-constant coefficient and deformed geometries, in: C. Canuto and A. Quateroni, eds., *Spectral and High Order Methods for Partial Differential Equations* (North-Holland, Amsterdam, 1989) 91.
- [23] G. Strang and G.J. Fix, *An Analysis of the Finite Element Method* (Prentice-Hall, Englewood Cliffs, NJ, 1973).
- [24] S.J. Sherwin, Development and Implementation of Two- and Three-Dimensional h-p Finite Element Methods on Unstructured Meshes, PhD Thesis, Princeton University, 1995.
- [25] C. Canuto and A. Quateroni, Approximation results for orthogonal polynomials in Sobolev spaces. *Math. Comput.* 38 (1982) 67.
- [26] C.W. Gear, *Numerical Initial Value Problems in Ordinary Differential Equations*. Prentice Hall, Englewood Cliffs, NJ, 1971.
- [27] G. Strang, *Linear Algebra and its Application*, 3rd edition (Harcourt Brace Jovanovich, Orlando, FL, 1986).
- [28] C. Canuto and A. Quateroni, Variational methods in the theoretical analysis of spectral approximations, in: R.G. Voight, D. Gottlieb and M.Y. Hussaini, eds., *Spectral Methods for Partial Differential Equations* (SIAM-CBMS, Philadelphia, PA, 1984) 55.
- [29] H.B. Moffat, Viscous and resistive eddies near a sharp corner, *J. Fluid Mech.* 18 (1964).
- [30] G.H. Wannier, A contribution to the hydrodynamics of lubrication. *Q. Appl. Math.* 8 (1950) 1.
- [31] L.I.G. Kovasznay, Laminar flow behind a two-dimensional grid, *Proc. Cambridge Philos. Soc.* (1948) 44.
- [32] M. Hammanche and M. Gharib, An experimental study of the parallel and oblique shedding in the wake from circular cylinders, *J. Fluid Mech.* 232 (1991) 567.
- [33] G.E. Karniadakis and G.S. Triantafyllou, Frequency selection and asymptotic states in laminar wakes, *J. Fluid Mech.* 199 (1989) 447.